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Fuzzy Spatial Algebra (FUSA): Formal Specification of Fuzzy Spatial Data Types and Operations for Databases and GIS

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Spatial database systems and Geographic Information Systems (GIS) are mainly able to support geographical applications that deal with *crisp spatial objects*, that is, objects whose extent, shape, and boundary are precisely determined. But geoscientists have pointed out for a long time that there is also a need to represent *fuzzy spatial objects* that reveal an intrinsically vague or blurred nature and structure and feature indeterminate boundaries and/or interiors. A spatial object is fuzzy if locations exist that cannot be assigned completely to the object or to its complement. In this article, we propose an abstract, formal, and conceptual type system called *Fuzzy Spatial Algebra* (*FUSA*) that provides a collection of *fuzzy spatial data types* for *fuzzy points*, *fuzzy lines*, and *fuzzy regions* in the two-dimensional Euclidean space. We introduce a set of expressive spatial operations such as *fuzzy union*, *fuzzy intersection*, and *fuzzy difference* to perform geometric computations on fuzzy spatial objects. As a specialty, users may exert influence on how spatial fuzziness is interpreted and handled in these operations. Our formal framework is based on fuzzy set theory and fuzzy topology. FUSA is designed to serve as a specification of its implementation in a spatial database and GIS context. We show the applicability of FUSA and its possible embedding into the query languages of extensible database systems by employing a running example.

CCS Concepts: • **Information systems → Geographic information systems**; *Database design and models*; • **Computing methodologies → Vagueness and fuzzy logic;**

Additional Key Words and Phrases: Fuzzy set, fuzzy logic, fuzzy spatial data type, spatial fuzziness, fuzzy point, fuzzy line, fuzzy region, fuzzy spatial operation

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1 Introduction

Representing, storing, querying, and manipulating spatial or geometric information is important for many advanced database applications. Specialized systems like Geographic Information Systems (GIS) and spatial database systems to a large extent provide the needed technology to support these applications. But so far, these systems have implicitly assumed that the extent, the interior, and the border of a spatial phenomenon are precisely determined, homogeneous, and universally recognized. From this perspective, spatial phenomena are typically represented by *points* with exactly known coordinates, *lines* linking a series of exactly known points, and *regions* bounded by exactly defined lines called *boundaries*. Special data types called *spatial data types* (see [67] for a survey) have been designed for modeling these spatial data. Due to their inherently complex structure, we speak of *spatial objects* as values of these data types. The properties of the space at the points, along the lines, and within the regions are given by attributes whose values are assumed to be constant over the total extent of the objects. Well-known examples are especially man-made spatial objects representing engineered artifacts, such as highways, houses, or bridges, and predominantly immaterial spatial objects exerting social control like countries, states, school districts, and land parcels with their political, administrative, and cadastral boundaries. We denote these kinds of entities as *crisp spatial objects*.

For a long time, geoscientists have increasingly realized that the current mapping of spatial phenomena of the real world to exclusively crisp spatial objects is an insufficient abstraction process for many spatial and geographical applications since the feature of *spatial fuzziness* is inherent to many geometric and geographic data. Spatial fuzziness captures the property of many spatial objects in reality that reveal an intrinsically vague or blurred nature and feature indeterminate boundaries and/or interiors; we call such entities *fuzzy spatial objects*. A spatial object is fuzzy if locations exist that cannot be assigned completely to the object or to its complement. Examples are natural, social, and cultural phenomena like land features with continuously changing properties (for example, population density, soil strata, soil quality, vegetation, pollution, temperature, air pressure), lakes, oceans, deserts, and English speaking areas. As another example, the transition between a valley and a mountain cannot be exactly ascertained since it is smooth and seamless and both objects are hence not crisp.

From the beginning, the deployment of fuzzy set theory has been a popular approach in geographic information science and the geosciences to modeling vague spatial data. Although the geosciences have demonstrated the usefulness of this approach to geographic application scenarios and some general concepts have been developed for the modeling of fuzzy spatial data, the authors sense a stagnation and lack of progress of this approach due to three main problems. First, available approaches reveal conceptual weaknesses such as the simplicity and restricted expressiveness of the allowed structures of fuzzy spatial data, the limited support of special data types for representing fuzzy spatial data, lacking closure properties of spatial operations on fuzzy spatial data resulting in invalid and/or degenerate results, and a limited functionality following from small sets of spatial operations only. Second, there is a lack of conceptual clarity of the characteristic features of fuzzy spatial data and their operations, including their formal definition and representation. Third, as a consequence, implementations for representing and processing fuzzy spatial data are rare.

In this article, we focus on solving the first and second problem and propose an abstract, formal, and conceptual type system called *Fuzzy Spatial Algebra* (*FUSA*) that enhances our understanding of fuzzy spatial concepts and provides a collection of *fuzzy spatial data types* for *fuzzy points*, *fuzzy lines*, and *fuzzy regions* in the two-dimensional Euclidean space. Further, we introduce a set of expressive spatial operations such as *fuzzy union*, *fuzzy intersection*, and *fuzzy difference* to perform geometric computations on fuzzy spatial objects. As a specialty, users may exert influence on how

spatial fuzziness is interpreted and handled in these operations. Due to space restrictions, we do not cover numerical operations, topological relationships, and directional relationships on fuzzy spatial objects. Our formal framework is based on fuzzy set theory and fuzzy topology. FUSA is designed to serve as a specification of its implementation in a spatial database and GIS context.

Section 2 discusses, classifies, and compares related work. Section 3 introduces our formal concept of fuzzy spatial data types in FUSA. Section 4 formally defines a variety of geometric operations on fuzzy spatial objects. Finally, Section 5 draws some conclusions and discusses future work.

2 Related Work

In this section, we discuss several different aspects of related work. Section 2.1 characterizes spatial fuzziness and distinguishes it from other kinds of spatial imperfection. Precursors of fuzzy spatial models with their many-valued logic include approaches based on a three-valued logic; they are sketched in Section 2.2. Section 2.3 provides examples of geoscience and GIS applications that leverage fuzzy set theory. Section 2.4 foregrounds the concept of fuzzy spatial data types and presents and compares available modeling approaches to fuzzy spatial data. Section 2.5 deals with proposed operations on fuzzy spatial data.

2.1 Spatial Fuzziness as an Occurrence of Spatial Imperfection

Imperfect information is ubiquitous in the real world. Its suitable treatment has a long history in many research fields and has been motivated by the objective to build systems that adequately model and reflect the real world. It has led to many attempts to achieve a classification of imperfection and to elucidate the relationships between them. We here follow the work by Parsons [59] which distinguishes five types of *imperfection* that may pervade data: incompleteness, inconsistency, uncertainty, imprecision, and vagueness. *Incompleteness* arises from the lack of relevant information and thus from the absence of a value. In a spatial context, an example consists in lacking radar signals yielding the positions of an aircraft or a ship. *Inconsistency* refers to the situation of two or more conflicting and ambiguous values for the same variable or entity that cannot be (easily) accommodated. In a spatial context, an example refers to different data sources that might result in diverging representations of the common border of two land properties. A possible solution to resolving inconsistency is to retract the data from the least reliable sources, provided that they can be determined.

Uncertainty, which is a largely overloaded term with different interpretations, is traditionally equated with randomness and chance occurrence. In a spatial context, it relates either to a lack of exact knowledge about the position and shape of a spatial object with an existing real boundary (*positional* uncertainty) or to the inability of measuring such an object precisely (*measurement* uncertainty). For example, geographic data derived from historical maps are often afflicted with positional inaccuracy and hence positional uncertainty. Probability theory is able to represent uncertainty and defines the membership grade of an entity in a set by a statistically defined probability function. It deals with the *expectation* of a future event, based on something known now.

Imprecision arises from the existence of a value that cannot be measured with sufficient accuracy. Intervals to set determinate limits to the actual values together with interval arithmetic are one method to express imprecision. Another method is rough set theory [61] that represents a set X by a pair of determinate lower and upper approximations. From the lower approximation we know that its elements belong definitely to X. The upper approximation is the set of elements that possibly belong to X. In a spatial context, an example is a lake that has its guaranteed smallest extent in dry periods and its possible largest extent in precipitation periods.

Vagueness (or *fuzziness*) denotes a special kind of imprecision. It is an intrinsic feature of a value or an object itself and refers to its inherent ambiguity, indeterminacy, bleariness, and ambivalence. Fuzzy set theory [86] is able to represent vagueness. It describes the *admission of the possibility* given by a *membership function* that an individual is a member of a set or that a given statement is true. The extent to which an element belongs to a set is given by a *membership degree*. Hence, vagueness is not based on the expectation of a future event; it results from the imprecision of the meaning of a concept. In a spatial context, *spatial vagueness* or *spatial fuzziness* describes the imprecision of a spatial object that certainly has an extent but inherently cannot or does not have a precisely definable interior and/or boundary. Examples of fuzzy spatial objects include mountains, valleys, biotopes, oceans, and many other geographic features whose interior cannot be unambiguously and completely assigned to them and/or which cannot be rigorously bounded by a sharp line. In the remainder of this article, we will exclusively deal with spatial fuzziness.

2.2 Approaches Leveraging Three-Valued Logic in GIS Applications

A few models can be regarded as precursors of the fuzzy approaches to representing vague spatial data. Although these models are different in detail, they have in common that they are based on a three-valued logic with the truth values *true*, *false*, and *maybe* (or any other intermediary truth value) and do not leverage a many-valued logic as the fuzzy approaches do. Examples are the *Egg-Yolk model* [24], *regions with broad boundaries* [22], *lines with broad boundaries and broad interiors* [21], the *Qualitative Min-Max model* [3], and the *Vague Spatial Algebra* (*VASA*) [60]. They all represent a vague spatial object by decomposing it into three spatial parts that correspond to the three truth values: (i) a spatial part that *definitely* belongs (*true*) to the object and is called yolk, minimum limit, or kernel in different models, (ii) a spatial part that *possibly* belongs (*maybe*) to the object and is called white, broad boundary, broad interior, maximum limit, or conjecture part by the different models, and (iii) a spatial part that does *definitely not* belong (*false*) to the object, is the complement of the aforementioned two spatial parts, and is called the exterior.

The representation of vague real-world phenomena by this three-part concept can be a limitation for spatial applications since often more levels of spatial fuzziness can be distinguished. Fuzzy set theory allows the representation of an arbitrary number of fuzziness levels that are either application defined or user-defined. In the remainder of this article, we will only deal with conceptual models that employ fuzzy set theory and consequently use more than three truth values to represent different levels of spatial fuzziness.

2.3 Approaches Leveraging Fuzzy Concepts in Geoscience and GIS Applications

Since the end of the eighties, geoscientists and GIS experts became increasingly interested in considering spatial fuzziness in geographic applications [6, 29, 37]. Geoscientists have focused on finding suitable geographic applications that can benefit from fuzzy set theory and on exploring application-specific membership functions (for example, [35, 57, 75]) that can adequately express spatial fuzziness. GIS experts have aimed to develop computational methods and systems that deploy fuzzy concepts.

The geosciences have made a large number of proposals how fuzzy set theory could benefit geographic applications. We can only mention a few examples here. Burrough [4], Lagacherie et al. [43], and Qi and Zhu [62] apply fuzzy set theory to soil profiles, soil classification, soil boundaries, choropleth soil maps, and land evaluation classifications. The usefulness of fuzzy set theory for modeling natural vague spatial objects and ways to choose suitable application-specific membership functions are described by Burrough [5]. Usery [77] provides a first definition of a *fuzzy feature* that discriminates between its *core* and its *boundary*. Tang et al. [76] perform change detection of land cover from satellite images. Land cover is regarded as a collection of fuzzy spatial objects,

and the degree of change is calculated using fuzzy reasoning. Liu and George [48] apply fuzzy sets and fuzzy clustering to mining weather data. Grubesic [32] uses the same principles for crime hot spot detection. Robinson [64] discusses issues and challenges of incorporating fuzzy sets in ecological modeling. Asadi et al. [2] develop a fuzzy model to represent the strength of rocks and rock masses. Jadidi et al. [35] propose a model based on fuzzy set theory to represent vague coastal risk zone boundaries. A number of membership functions are introduced to represent different risk parameters. A spatial decision support system that incorporates spatial fuzziness and fuzzy spatial objects is introduced by Carniel and Schneider [11], Carniel et al. [16], and Carniel et al. [8]. Khalfi et al. [40] focus on how fuzzy spatial objects can be stored in document-oriented NoSQL databases in the context of building fuzzy spatial big data applications. Two questions that remain unanswered in these publications is, first, how the described fuzzy spatial features can be formally defined and, second, how they can be mapped to a computer representation.

The GIS field has aimed to develop computational solutions to integrate fuzzy concepts into GIS. From the numerous publications we can only present a few examples here. Sui [74] shows that incorporating fuzzy set theory into GIS modeling can provide more details about the gradual transition of urban land value. Altman [1] introduces the concept of grid-based fuzzy regions in which each grid cell is labeled with a membership degree. Wang and Hall [82] discuss fuzzy representations of geographical boundaries in GIS. Leung and Yan [44] present simple concepts for fuzzy points and fuzzy regions, specify "concentric" membership functions with decreasing membership degrees emanating from the core of a fuzzy spatial object, and deal with the fuzzy-point-in-fuzzy-polygon test in GIS. Oh and Jeong [57] show the usefulness of the fuzzy set approach in GIS for evaluating urban residential environments. Robinson [63] shows the usefulness of fuzzy set theory in GIS to represent imprecision and discusses how concepts such as membership functions and alpha-cuts (see Sections 2.5 and 3.2.1) could be leveraged. The book by Lodwick [49] presents methods, algorithms, and applications to model and analyze surfaces derived from fuzzy spatial data in GIS. Semeraro et al. [72] introduce a fuzzy expert system integrated into a GIS environment for an effective management of natural protected areas in order to mitigate their fire vulnerability. Rodríguez et al. [65] present a GIS-based approach to identifying the best sites to install a bioenergy plant. Guo and Du [33] propose a GIS-based method for building fuzzy regions from point sets. Carniel and Schneider [15] present a survey of fuzzy approaches in the context of spatial data science. Noble et al. [54] propose an approach based on fuzzy logic to model decision-making maps for assessing and designing marine protected areas.

The main problem of the aforementioned approaches is the lack of a clear and consistent concept (and implementation) how to represent fuzzy spatial data in GIS. The main difference to our approach is that we propagate fuzzy spatial data types as a solution to this problem. We will discuss further GIS related literature that focuses on fuzzy spatial data types in Sections 2.4 and 2.5.

2.4 Approaches to Modeling Fuzzy Spatial Data

There is a broad consensus in the spatial database and GIS communities that special data types, called *spatial data types*, are needed to suitably represent geometric data in the respective systems. Spatial data types for point, line, and region objects provide a fundamental abstraction for modeling the geometric structure of objects in the (here two-dimensional) space as well as their properties, operations, and predicates. While data types for crisp, exact spatial objects have been widely explored from both a formal definition and implementation point of view, this is not the case for *fuzzy spatial data types* (Section 3). In this subsection, we present, group, and compare the available approaches, identify their weaknesses and strengths, and provide an overview of them in Table 1.

Our comparison criteria are threefold. First, we check whether the available approaches offer a variety of fuzzy spatial data types to represent fuzzy spatial objects of different dimensions, such as

Table 1. Comparison of Approaches to Modeling Fuzzy Spatial Data Types

Approaches	Comparison Criteria: Data Type Features							
	Kinds of Fuzzy Spatial Data Types				Geometric Structure		Membership Cardinality	
	fuzzy point	fuzzy line	fuzzy region	fuzzy region with holes	simple	complex	finite	infinite
Reference group 1[a]			✓		✓		✓	
Reference group 2[b]			✓		✓			✓
Zhan [87]			✓		✓	✓	✓	
Reference group 3[c]			✓	✓	✓			✓
Reference group 4[d]			✓	✓	✓	✓		✓
Verstraete et al. [81]	✓		✓		✓			✓
Schneider [69]		✓	✓	✓	✓	✓		✓
Reference group 4[e]	✓	✓	✓	✓	✓			✓
Reference group 5[f]	✓	✓	✓	✓	✓	✓		✓

[a]includes Krishnapuram et al. [42] and Schmitz and Morris [66].
[b]includes Wang and Hall [82], Molenaar and Cheng [52], Cheng [20], Yu and Liu [85], Verstraete [78], Verstraete [79], and Zheng et al. [88].
[c]includes Goodchild et al. [30], Verstraete [80], and Jadidi et al. [35].
[d]includes Carniel et al. [17], Jifa and Tiejun [38], and Carniel and Schneider [10].
[e]includes Liu and Shi [45], Liu and Shi [46], and Tang [75].
[f]includes Schneider [68], Dilo [27], Dilo et al. [28], Carniel and Schneider [9], and FUSA.

fuzzy points, *fuzzy lines*, and *fuzzy regions*. Second, we determine whether the approaches support *simple* fuzzy spatial data types only or even *complex* data types. Simple fuzzy spatial data types only provide simple object structures like single-component fuzzy points, fuzzy lines, and fuzzy regions possibly with holes. However, from an application perspective, it is well known that they are insufficient to cope with the variety and complexity of geographic reality. Further, from a formal perspective, they are not closed under the geometric operations *fuzzy intersection*, *fuzzy union*, and *fuzzy difference*. Complex fuzzy spatial data types solve these problems. They provide universal and versatile fuzzy spatial objects with finitely many components, permit fuzzy regions with holes, and are closed under fuzzy geometric operations (Section 4). Third, we determine whether an approach is able to discriminate a finite or infinite number of fuzziness levels in its underlying model. This is decided by the cardinality of the possible membership degrees that can be assigned to each point of a fuzzy spatial object. If the cardinality is finite, the respective fuzzy spatial objects are unable to represent continuous transitions of the membership degrees of their underlying point sets. But smooth transitions are typical for vague real-world phenomena.

Table 1 lists both single approaches with unique comparison features as well as reference groups with equal and unique comparison features and sorts them according to increasing level of expressiveness. We sort the members of each reference group increasingly by their publication year. With regard to the first criterion, the majority of the approaches (see the first five rows in Table 1) focus on the definition of fuzzy regions only. This is motivated by the importance of areal objects and the interest in applying spatial operations on them. The remaining approaches aim at more complete type systems and include fuzzy points and fuzzy lines together with respective operations on them. With regard to the second criterion, all approaches support simple object structures, and only some of them permit complex object structures. Those approaches that only support simple object structures are limited in their expressiveness and thus usability, provide operations of restricted functionality only, and are unable to satisfy closure properties.

Table 2. Comparison of Approaches to Modeling Fuzzy Spatial Operations

Approaches	Comparison Criteria: Operation Categories and Features						
	Operations Returning Crisp Spatial Objects	Geometric Set Operations Returning Fuzzy Spatial Objects			Type-Dependent Geometric Operations	Membership-based Operations	Closure Properties
		standard	t-(co)norm	diff.			
Reference group 1[a]	✓						
Zheng et al. [88]	✓						
Verstraete [80]	✓					✓	
Verstraete [79]		✓					✓
Verstraete [78]	✓						
Jifa and Tiejun [38]	✓					✓	
Schneider [68]	✓	✓					✓
Reference group 2[b]	✓	✓					✓
Verstraete et al. [81]	✓	✓	✓		✓		✓
FUSA	✓	✓	✓	✓	✓	✓	✓

[a]includes Zhan [87], Tang [75], and Schmitz and Morris [66].
[b]includes Dilo [27] and Dilo et al. [28].

With regard to the third criterion, only reference group 1 and Zhan [87] consider a finite cardinality of membership degrees since they focus on implementations aspects and hence their membership functions are based on finite alpha-cuts (see Section 3.2.1). All other approaches allow infinite membership cardinality and are able to model smooth transitions of membership degrees. An important goal of FUSA is to unify and generalize the properties of all available approaches and formally specify general *fuzzy spatial data types* for *complex fuzzy point*, *fuzzy line*, and *fuzzy region* objects that comprise *simple* objects as a special case, permit multiple components and regions with holes, and rest on an *infinite cardinality of membership degrees*.

Furthermore, a few approaches represent lower-dimensional fuzzy spatial objects by higher-dimensional ones. For example, Verstraete et al. [81] specify fuzzy point objects as indeterminate areal geometries that we would regard and represent as fuzzy region objects. We consider this as a "break of dimensional abstraction". Some further approaches [27, 28, 75] model fuzzy spatial objects as mixed-dimensional entities. For example, a polluted river is modeled as a composition of a fuzzy line object and a fuzzy region object. This leads to heterogeneous fuzzy spatial data types, that is, to fuzzy geometry collections [14]. But these are compositions of the fuzzy spatial data types defined in FUSA and can therefore be derived from them if this should be desired.

2.5 Approaches to Operations on Fuzzy Spatial Data

In this subsection, we explore the approaches from Table 1 regarding their support of spatial operations on fuzzy spatial objects as input and provide an overview and grouping of them in Table 2. We distinguish geometric operations returning crisp spatial objects, geometric set operations returning fuzzy spatial objects, type-dependent geometric operations, and membership-based geometric operations. Further, we investigate whether the geometric operations of these approaches fulfill important closure properties.

Geometric operations returning crisp spatial objects transform a fuzzy spatial object completely or partially into a crisp spatial object. An example is the computation of the *core* of a fuzzy spatial object that collects all points with the membership degree 1 into a crisp spatial object. Similarly, the *support* of a fuzzy spatial object is an operation that collects all points with membership degree greater than 0 into a crisp spatial object. Another example is the computation of a crisp spatial

alpha-cut object that contains all points with a membership degree that is larger than (or equal to) a given threshold membership degree. Almost all approaches include at least one such operation. The most common operations are the operations *support* and *core*, which are generalized from fuzzy set theory to be applicable to fuzzy spatial objects.

Geometric set operations returning fuzzy spatial objects calculate the *geometric intersection*, *geometric union*, and *geometric difference* of two fuzzy spatial objects, respectively. Usually, standard membership transformation functions calculate the resulting membership degrees for each geometric set operation such as the *minimum* function for geometric intersection, the *maximum* function for geometric union, and the standard fuzzy difference function (that is, the standard intersection between the first element and the complement of the second element) for geometric difference. However, applications can require other context-dependent functions such as *triangular norms* (*t-norms*) for geometric intersection, *triangular conorms* (*t-conorms*) for geometric union, and *bounded difference* and *symmetric difference* for geometric difference (see Section 4). Only a few approaches define geometric set operations with the standard membership transformations. Verstraete et al. [81] even allow the use of *t*-norms and *t*-conorms to calculate the membership degrees of the fuzzy intersection and union operations, respectively; advanced membership transformations for the fuzzy difference operation are not defined.

Type-dependent geometric operations can only be applied to particular fuzzy spatial data types. Examples are operations that determine the common points of two fuzzy line objects and the convex hull of a set of fuzzy points. Despite the meaningfulness of this category of operations, only Verstraete et al. [81] offer an example of such an operation, namely the convex hull operation on fuzzy regions.

Membership-based operations only process the membership degrees of a fuzzy spatial object. Examples are operations that extract the maximum membership degree of a fuzzy spatial object or allow to increase or decrease its membership degrees and thus change the meaning of its spatial fuzziness. Current approaches hardly support this category of operations. They mainly allow the determination of the maximum membership degree of a fuzzy spatial object.

Closure properties ensure that the result of a fuzzy spatial operation leads to a valid and consistent object of a fuzzy spatial data type. They, for example, exclude dangling points and punctures in fuzzy spatial objects. Only the approaches that offer geometric set operations also guarantee closure properties.

In Table 1 we have listed a number of references that do not appear in Table 2. These references deal with *fuzzy topological relationships* (Wang and Hall [82], Goodchild et al. [30], Yu and Liu [85], Liu and Shi [45], Liu and Shi [46], Carniel et al. [17], Jadidi et al. [35], Carniel and Schneider [9], Carniel and Schneider [10]) and *numerical operations* (Krishnapuram et al. [42], Molenaar and Cheng [52], Schneider [69], Cheng [20]) on fuzzy spatial objects. We plan to cover these two important kinds of operations in a future article.

The goal of FUSA is to formally define a comprehensive superset of expressive fuzzy geometric operations, which are based on solid, sound, and general properties and definitions, in all categories. However, we do not aim at developing a type system with a "complete" set of data types and operations. It is common consensus in the spatial database and GIS communities that this is impossible since always new data types, operations, and predicates can be required and designed. It is then more favorable to retroactively add them to the algebra; hence, we take an extensible approach.

3 Fuzzy Spatial Data Types

This section introduces our fundamental formal definitions of *fuzzy spatial data types* for *fuzzy points* (Section 3.5), *fuzzy lines* (Section 3.6), and *fuzzy regions* (Section 3.7). But first we present

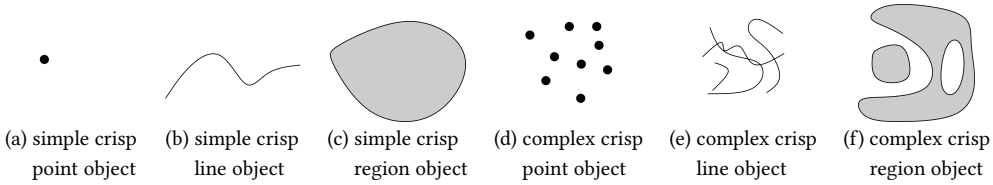


Fig. 1. Examples of crisp spatial objects.

some foundations for these definitions. Section 3.1 provides an intuitive conception and comparison of crisp and fuzzy spatial objects. Section 3.2 presents some notions from fuzzy set theory and fuzzy set topology needed for the formal definition of fuzzy spatial objects. In Section 3.3 we define the fuzzy topological space $(\mathbb{R}^n, \mathcal{T}_n^u)$ as the basis of the fuzzy line and fuzzy region data types. Finally, Section 3.4 introduces a running application example to illustrate and apply the concepts of FUSA in the later subsections.

3.1 Crisp Spatial Objects versus Fuzzy Spatial Objects

As indicated in Section 1, nowadays spatial data handling as well as corresponding software systems such as spatial databases and GIS almost exclusively process *crisp* spatial objects. These are characterized by precisely determined, exact, and homogeneous geometries in the Euclidean plane and have been standardized by means of *spatial data types* [55, 56] for two-dimensional *point*, *line*, and *region* objects. The assumption of crisp boundaries and homogeneous interiors of spatial objects harmonizes very well with their internal computer representation and processing favoring precise and unique data structures and algorithms. Examples of spatial objects are the locations of landmarks, lighthouses, hydrants, power poles, traffic signs, and ATMs represented as crisp point objects, the routes of highways, streets, guardrails, and rivers modeled as crisp line objects, and the extents of countries, cities, districts, buildings, houses, and airports shaped as crisp region objects. Figure 1 illustrates the crisp geometries of such objects and distinguishes simple and complex object structures, the former being a special case of the latter.

While man-made spatial objects nicely accommodate with their modeling as crisp entities, many natural, social, and scientific phenomena reveal an inherently imprecise, blurred, vague, and fuzzy nature but have so far been artificially pressed into a crisp representation and modeling scheme. For a long time and in a number of geospatial applications (see Section 2.3), geoscientists have shown the need of and their interest in representing spatial vagueness due to the inadequacy of crisp modeling techniques. They have selected fuzzy set theory as a promising and popular approach. But recent years have revealed a stagnation of progress in this direction. The authors see the following main reasons for this: (i) There is a lack of conceptual clarity of the essence, structure, and characteristic features of fuzzy spatial data and their operations, including their formal definition and representation. (ii) There is only limited support of special data types for representing fuzzy spatial data and their operations. (iii) As a consequence, implementations for representing and processing fuzzy spatial data as well as systems including them are rare.

In this article, we introduce and formally define the *Fuzzy Spatial Algebra* (*FUSA*). It provides *fuzzy spatial data types* for representing *fuzzy point objects* (Figures 2(a) and 2(d)), *fuzzy line objects* (Figures 2(b) and 2(e)), and *fuzzy region objects* (Figures 2(c) and 2(f)), which can have either simple or complex structures. In addition, FUSA includes *fuzzy spatial operations* defined on these data types for processing *fuzzy spatial objects*. FUSA fulfills all the data type features discussed in Section 2.4 since it comprises all three kinds of fuzzy spatial data types, allows both simple and complex geometric structures, and uses the infinite interval $[0, 1]$ as its range of membership

(a) simple fuzzy point object (b) simple fuzzy line object (c) simple fuzzy region object (d) complex fuzzy point object (e) complex fuzzy line object (f) complex fuzzy region object

Fig. 2. Examples of fuzzy spatial objects. Darker parts denote higher membership degrees than lighter parts.

degrees (see Sections 3.5 to 3.7). It also includes all operation categories and features discussed in Section 2.5 (see Section 4).

Our modeling strategy is that in the same manner as crisp sets are generalized to fuzzy sets, FUSA generalizes crisp spatial objects to fuzzy spatial objects. This means it maintains the geometric structure of point, line, and region objects but allows partial membership of single points in these objects. In this sense, a fuzzy point object represents a set of blurred zero-dimensional points by tagging each point with a membership degree. Examples are drug trafficking locations in crime analysis applications, the sites of terror suspects in homeland security applications, the origins of disease outbreaks in disaster management applications, and the positions of prehistoric sites in archaeological applications. A fuzzy line object is a set of indefinite one-dimensional, curvilinear structures by labeling each point with a membership degree. Examples are animal routes in wildlife habitat applications, traces of river pollution in ecological applications, terrorist escape routes in homeland security applications, and the condition of asphalt on streets in urban planning applications. A fuzzy region object represents a set of two-dimensional vague areas by associating each point with a membership degree. Examples are endangered species in wildlife conservation applications, climate change and sea level rise in coastal zone management applications, soil strata in agricultural applications, air polluted areas from emissions of coal-fired power plants and industrial parks in ecological applications, terrorist and criminal activities in homeland security applications, and evacuation zones in disaster management applications.

3.2 Basic Concepts from General and Fuzzy Set Theory and Topology

In this subsection, we summarize some basic formal concepts of *fuzzy sets* (Section 3.2.1), *fuzzy set operations* (Section 3.2.2), and *fuzzy topology* (Section 3.2.3). We briefly sketch that their crisp counterparts form special cases. Since we are especially interested in fuzzy spatial objects with "smooth" membership functions, we also present some basic concepts of *semi-continuous* and *continuous real-valued functions* (Section 3.2.4).

3.2.1 Crisp Sets versus Fuzzy Sets. Fuzzy set theory [86] is an extension and generalization of classical Boolean set theory. Let X be a classical (crisp) set of objects, called the *universe* (*of discourse*). Membership in a classical subset A of X is then given by the *characteristic function* $\chi_A : X \longrightarrow \{0, 1\}$ such that for all $x \in X$ holds that $\chi_A(x) = 1$ if, and only if, $x \in A$, and $\chi_A(x) = 0$ otherwise. A fuzzy set relaxes this strict discrimination and maps each element of X to a value of the real interval $[0, 1]$ indicating the *degree of membership* of the element in the set in question. Larger (smaller) values designate higher (lower) degrees of set membership. Let X again be the universe. Then the function $\mu_{\tilde{A}} : X \longrightarrow [0, 1]$ is called the *membership function* of the *fuzzy set* $\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) \mid x \in X\}$. All elements in X receive a valuation regarding their membership in $\tilde{A}$. Hence, fuzzy set theory permits that an element has partial membership in a fuzzy set as well as multiple and different *membership degrees* in different fuzzy sets. Those elements $x \in X$ that in the classical sense do not belong to $\tilde{A}$ get the membership degree $\mu_{\tilde{A}}(x) = 0$; elements $x \in X$ that completely belong to $\tilde{A}$ get the membership degree $\mu_{\tilde{A}}(x) = 1$.

Table 3. Definitions of Fuzzy Set Operations (Let $\tilde{A}$ and $\tilde{B}$ be Fuzzy Sets in X)

Fuzzy Set Operation	Definition
fuzzy complement	$\neg\tilde{A} = \{(x, \mu_{\neg\tilde{A}}(x)) \mid x \in X \wedge \mu_{\neg\tilde{A}}(x) = 1 - \mu_{\tilde{A}}(x)\}$
fuzzy set intersection	$\tilde{A} \cap \tilde{B} = \{(x, \mu_{\tilde{A}\cap\tilde{B}}(x)) \mid x \in X \wedge \mu_{\tilde{A}\cap\tilde{B}}(x) = \min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))\}$
fuzzy set union	$\tilde{A} \cup \tilde{B} = \{(x, \mu_{\tilde{A}\cup\tilde{B}}(x)) \mid x \in X \wedge \mu_{\tilde{A}\cup\tilde{B}}(x) = \max(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))\}$
fuzzy set difference	$\tilde{A} - \tilde{B} = \tilde{A} \cap \neg\tilde{B} = \{(x, \mu_{\tilde{A}-\tilde{B}}(x)) \mid x \in X \wedge \mu_{\tilde{A}-\tilde{B}}(x) = \min(\mu_{\tilde{A}}(x), 1 - \mu_{\tilde{B}}(x))\}$
fuzzy bounded difference	$\tilde{A} \dot{-} \tilde{B} = \{(x, \mu_{\tilde{A}\dot{-}\tilde{B}}(x)) \mid x \in X \wedge \mu_{\tilde{A}\dot{-}\tilde{B}}(x) = \max(0, \mu_{\tilde{A}}(x) - \mu_{\tilde{B}}(x))\}$
fuzzy symmetric difference	$\tilde{A} \,\Delta\, \tilde{B} = (\tilde{A} - \tilde{B}) \cup (\tilde{B} - \tilde{A})$
fuzzy absolute difference	$\tilde{A} \mid\!-\!\mid \tilde{B} = \{(x, \mu_{\tilde{A}\mid\!-\!\mid\tilde{B}}(x)) \mid x \in X \wedge \mu_{\tilde{A}\mid\!-\!\mid\tilde{B}}(x) = \lvert\mu_{\tilde{A}}(x) - \mu_{\tilde{B}}(x)\rvert\}$
fuzzy set containment	$\tilde{A} \subseteq \tilde{B} \Leftrightarrow \forall x \in X : \mu_{\tilde{A}}(x) \leq \mu_{\tilde{B}}(x)$

3.2.2 Crisp Set Operations versus Fuzzy Set Operations. Fuzzy set operations and fuzzy set inclusion generalize the corresponding Boolean operations. We comply with their definitions in [86] and will consider their generalization in terms of t-norms and t-conorms in Section 4.3.1. Let $\tilde{A}$ and $\tilde{B}$ be fuzzy sets in X. Table 3 presents the formal definitions of fuzzy set operations. In addition, proper fuzzy set containment ($\subset$) and fuzzy set equality ($=$) are defined correspondingly. If $\tilde{A}$ and $\tilde{B}$ are crisp sets A and B, their elements only have membership degrees 1 and 0, and their set operations result in special cases of the fuzzy set and containment operations. In this case, the fuzzy set difference and the fuzzy bounded difference provide the same result as the classical difference, while the fuzzy symmetric difference and the fuzzy absolute difference provide the same result as the classical symmetric difference.

Three further operations named *α-cut*, *support*, and *core* map fuzzy sets to crisp sets. Let $\tilde{A}$ be a fuzzy set. A [*strict*] *α-cut* A_α [A^*_α] of $\tilde{A}$ for a specified value α is the crisp set $A_\alpha[A^*_\alpha] = \{x \in X \mid \mu_{\tilde{A}}(x) \geq [>]\ \alpha \ \wedge\ 0 \leq \alpha \leq [<]\ 1\}$. The *support* of $\tilde{A}$, denoted by $supp(\tilde{A})$, is the crisp set A^*_0, that is, it contains those elements that contribute to $\tilde{A}$ to some degree. The *core* of $\tilde{A}$ contains its crisp points, that is, $core(\tilde{A}) = A_1$. $\tilde{A}$ is called *normal* if its core is not empty. For a fuzzy set $\tilde{A}$ and $\alpha, \beta \in [0,1]$ holds that (i) $X = A_0$ and (ii) $\alpha < \beta \Rightarrow A_\alpha \supseteq A_\beta$. The set of all distinct levels $\alpha \in [0,1]$ of $\tilde{A}$ is called the *level set* $\Lambda_{\tilde{A}}$ of $\tilde{A}$ and is defined as $\Lambda_{\tilde{A}} = \{\alpha \in [0,1] \mid \exists x \in X : \mu_{\tilde{A}}(x) = \alpha\}$.

3.2.3 General Topology versus Fuzzy Topology. Fuzzy topology [19, 47] is a straightforward extension and generalization of *general topology* [53]. It enables the distinction of different fuzzy topological structures, such as the closure and the interior of a fuzzy set $\tilde{A}$. Let $[0,1]^X = \{f \mid f : X \longrightarrow [0,1]\}$ be the set of all mappings from a non-empty crisp set X to the real interval $[0,1]$. Let 1^X and 0^X be the functions that correspond to the whole set X and the empty set $\emptyset$, respectively. A *fuzzy topology* on X is a family $\mathcal{T} \subseteq [0,1]^X$ of fuzzy sets that satisfies the following conditions: (i) $1^X \in \mathcal{T}$, $0^X \in \mathcal{T}$, (ii) $\tilde{A} \in \mathcal{T}, \tilde{B} \in \mathcal{T} \Rightarrow \tilde{A} \cap \tilde{B} \in \mathcal{T}$, and (iii) $\mathcal{S} \subseteq \mathcal{T} \Rightarrow \bigcup_{\tilde{A}\in\mathcal{S}} \tilde{A} \in \mathcal{T}$. By reducing the set $[0,1]^X$ to the set $\{0,1\}^X$ of mappings, we obtain the concept of general topology.

The pair $(X, \mathcal{T})$ is said to be a *fuzzy topological space*, and the elements of $\mathcal{T}$ are called *open fuzzy sets*. The family $\mathcal{T}'$ of all *closed fuzzy sets* in a fuzzy topological space $(X, \mathcal{T})$ is given by $\mathcal{T}' = \{\neg\tilde{A} \mid \tilde{A} \in \mathcal{T}\}$. The *closure* of a fuzzy set $\tilde{A}$ in a fuzzy topological space $(X, \mathcal{T})$ is the smallest closed fuzzy set containing $\tilde{A}$, that is, $cl_{\mathcal{T}}(\tilde{A}) = \bigcap\{\tilde{B} \mid \tilde{B} \in \mathcal{T}' \wedge \tilde{A} \subseteq \tilde{B}\}$. The *interior* of a fuzzy set $\tilde{A}$ in a fuzzy topological space $(X, \mathcal{T})$ is the largest open fuzzy set contained in $\tilde{A}$, that is, $int_{\mathcal{T}}(\tilde{A}) = \bigcup\{\tilde{B} \mid \tilde{B} \in \mathcal{T} \wedge \tilde{B} \subseteq \tilde{A}\}$. Considering a general topology T instead of a fuzzy topology $\mathcal{T}$ leads to the concept of a *general topological space* (X, T).

In a fuzzy topological space $(X, \mathcal{T})$ a fuzzy set $\tilde{U}$ in $\mathcal{T}$ is a *neighborhood* of a fuzzy set $\tilde{A}$ in $\mathcal{T}$ if, and only if, there exists an open fuzzy set $\tilde{V}$ in $\mathcal{T}$ such that $\tilde{A} \subseteq \tilde{V} \subseteq \tilde{U}$ holds. A similar definition can be given in a crisp topological space.

Sets in $\mathcal{T}$ (and T) can have irregularities such as missing or isolated elements. *Regularization* is a method to remove such irregularities. The fuzzy set $\tilde{A}$ is *regular closed* if, and only if, it is equal to the closure of its interior, that is, $\tilde{A} = cl_{\mathcal{T}}(int_{\mathcal{T}}(\tilde{A}))$. The fuzzy set $\tilde{A}$ is *regular open* if, and only if, it is equal to the interior of its closure, that is, $\tilde{A} = int_{\mathcal{T}}(cl_{\mathcal{T}}(\tilde{A}))$. These concepts can be specialized to crisp sets in general topological spaces. In Section 3.3, we will use these concepts in a spatial context.

To provide notions of the borders of fuzzy spatial objects, two concepts are used: *fuzzy boundary* [83] and *fuzzy frontier* [25]. Applying their definitions to crisp sets leads to the corresponding boundary definition from general topology. Formally, the *fuzzy boundary* of a fuzzy set $\tilde{A}$ in a fuzzy topological space $(X, \mathcal{T})$ is defined as the infimum of all closed fuzzy sets of $\mathcal{T}'$ in X with the property that $\forall \tilde{S} \in \mathcal{T}' \; \forall x \in X : \mu_{\tilde{S}}(x) \geq \mu_{cl_{\mathcal{T}}(\tilde{A})}(x)$ for which $\mu_{cl_{\mathcal{T}}(\tilde{A}) \cap cl_{\mathcal{T}}(\neg\tilde{A})}(x) > 0$ holds. Intuitively, the fuzzy boundary of a fuzzy set $\tilde{A}$ is the set of all elements of $\tilde{A}$ whose neighborhoods contain elements from both $\tilde{A}$ and its complement. Thus, it is composed of two parts: (i) the uncertain part of $\tilde{A}$ (that is, all elements with membership in $]0, 1[$), and (ii) the crisp boundary of the core of $\tilde{A}$. The first part is obtained from the intersection between $\tilde{A}$ and its complement. The second part is derived from the intersection of $\tilde{A}$ and the crisp part produced by the closure of the complement of $\tilde{A}$ (that is, the boundary of the core of $\tilde{A}$).

The *fuzzy frontier* of $\tilde{A}$ is a subset of and thus stricter than the fuzzy boundary of $\tilde{A}$. It is defined as the infimum of all closed fuzzy sets of $\mathcal{T}'$ in X with the property that $\forall \tilde{S} \in \mathcal{T}' \; \forall x \in X : \mu_{\tilde{S}}(x) \geq \mu_{cl_{\mathcal{T}}(\tilde{A})}(x)$ for which $\mu_{cl_{\mathcal{T}}(\tilde{A})}(x) - \mu_{int_{\mathcal{T}}(\tilde{A})}(x) > 0$ holds. Intuitively, the fuzzy frontier of $\tilde{A}$ encompasses all the elements belonging to the boundary of the support of $\tilde{A}$ annotated with the membership degree of the corresponding elements of $\tilde{A}$.

A fuzzy set $\tilde{A}$ is *bounded* [86] if A_α is bounded for all $\alpha \in \Lambda_{\tilde{A}}$. Let $f : P \to R$ be a function and $S \subset P$. Then $f|_S : S \to R$ is a function such that $f|_S(x) = \{f(x) \mid x \in S\}$. Let $(X, \mathcal{T})$ be a fuzzy topological space, and let $Y \subset X$. Then we call the family $\mathcal{U} = \{\mu_{\tilde{A}}|_Y \mid \tilde{A} \in \mathcal{T}\}$, which is a fuzzy topology for Y, the *relative fuzzy topology* of $\mathcal{T}$ to Y. Such a fuzzy topological space $(Y, \mathcal{U})$ is called a *subspace* of $(X, \mathcal{T})$. Two fuzzy sets $\tilde{A} \neq 0^X$ and $\tilde{B} \neq 0^X$ in $(X, \mathcal{T})$ are said to be *Q-separated* [58] if $cl_{\mathcal{T}}(\tilde{A}) \cap \tilde{B} = 0^X = \tilde{A} \cap cl_{\mathcal{T}}(\tilde{B})$. A fuzzy set $\tilde{A}$ in $(X, \mathcal{T})$ is called *disconnected* [58] if there exist two fuzzy sets $\tilde{B} \neq 0^X$ and $\tilde{C} \neq 0^X$ in the subspace $(A_0^*, \mathcal{U})$ such that $\tilde{B}$ and $\tilde{C}$ are Q-separated and $\tilde{A} = \tilde{B} \cup \tilde{C}$. A fuzzy set $\tilde{A}$ is *connected* [58] if it is not disconnected. Similar definitions hold for crisp sets.

3.2.4 Semi-continuous and Continuous Real-Valued Functions. The concept of semi-continuous and continuous real-valued functions will be applied in Section 3.3 to represent the fuzzy spatial nature of real-world phenomena. Suppose (X, T) is a general topological space, $x_0 \in X$, and $f : X \to \mathbb{R} \cup \{-\infty, +\infty\}$ is an extended real-valued function. The function f is called *upper semi-continuous* at x_0 if for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x) \leq f(x_0) + \epsilon$ for all $x \in U$ with $f(x_0) > -\infty$, and $f(x)$ tends to $-\infty$ as x tends towards x_0 with $f(x_0) = -\infty$. If X is a metric space (such as $\mathbb{R}^2$), this can be expressed as $\limsup_{x \to x_0} f(x) \leq f(x_0)$ where lim sup is the *limit superior* of the function f at element x_0. The function f is called upper semi-continuous if it is upper semi-continuous at every element $x \in X$. The function f is upper semi-continuous if, and only if, $\{x \in X \mid f(x) < \delta\}$ is an *open set* for every $\delta \in \mathbb{R}$.

The function f is called *lower semi-continuous* at x_0 if for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x) \geq f(x_0) - \epsilon$ for all $x \in U$ with $f(x_0) < +\infty$, and $f(x)$ tends to $+\infty$ as x tends

towards x_0 with $f(x_0) = +\infty$. If X is a metric space, this can be expressed as $\liminf_{x \to x_0} f(x) \geq f(x_0)$ where lim inf is the *limit inferior* of the function f at element x_0. The function f is called lower semi-continuous if it is lower semi-continuous at every element $x \in X$. The function f is lower semi-continuous if, and only if, $\{x \in X \mid f(x) > \delta\}$ is an *open set* for every $\delta \in \mathbb{R}$, or, alternatively, if, and only if, $\{x \in X \mid f(x) \leq \delta\}$ is a *closed set* for every $\delta \in \mathbb{R}$.

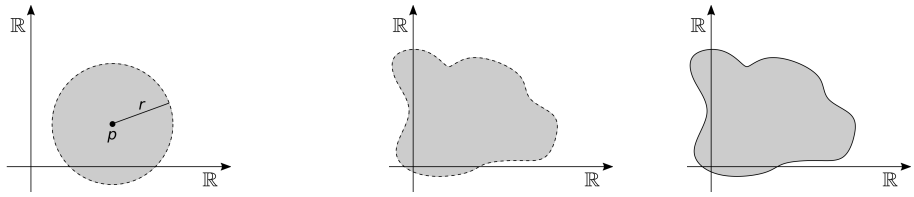
If a function f is upper semi-continuous or lower semi-continuous (but not both) at an element $x_0 \in X$, we call x_0 an *element of discontinuity*. It can be shown that f has a countable (that is, finite or countably infinite) set of discontinuities, called an F_σ *set* in the literature. Function f is called *finitely upper* (*lower*) *semi-continuous* if, and only if, f is upper (lower) semi-continuous and the number of elements of discontinuity is finite. In this case, we call the finite set of discontinuities an F_σ^{fin} set. This specialization is needed since geospatial applications only deal with fuzzy spatial objects that include finitely many discontinuities.

A function f is *continuous* at x_0 if, and only if, it is upper semi-continuous and lower semi-continuous at x_0. A function f is continuous if, and only if, it is upper semi-continuous and lower semi-continuous at every element $x \in X$.

A *piecewise* function f is a function defined by multiple subfunctions, each subfunction applied to a disjoint subdomain of the main function's domain. If the number of subfunctions is finite, we call f a *finitely piecewise* function. A (finitely) piecewise function is (finitely) (semi-)continuous on a given domain if (i) it is defined throughout that domain, (ii) its constituent functions are (finitely) (semi-)continuous on the corresponding subdomains, and (iii) there is no discontinuity at each endpoint of the disjoint subdomains within that domain.

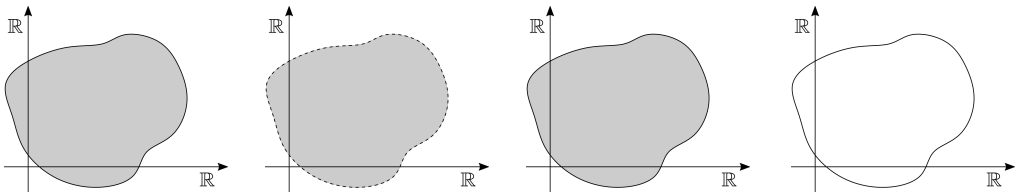
3.3 The Fuzzy Topological Space $(\mathbb{R}^n, \mathcal{T}_n^u)$

In this section, we deploy the concepts presented in Section 3.2 for the definition of the fuzzy topological spaces that form the basis of our fuzzy line and fuzzy region data types. Since we employ the Euclidean spaces $\mathbb{R}$ and $\mathbb{R}^2$ to define these data types (see Sections 3.6 and 3.7), the following definitions assume that $X = \mathbb{R}^n$ with $n \in \{1, 2\}$. Actually, most of the definitions are valid for $n > 2$. Different topologies can be defined on $\mathbb{R}^n$. For example, the trivial topology only contains the empty set $\emptyset$ and the whole set $\mathbb{R}^n$. For the definition of our topology of interest we leverage the fact that the set $\mathbb{R}$ of real numbers has the algebraic structure of an *ordered field* including the usual operations of addition, subtraction, and multiplication as well as an order relation "<". Let $p = (x_1, \ldots, x_n) \in \mathbb{R}^n$, $r \in \mathbb{R}^+$ with $\mathbb{R}^+ = \{x \in \mathbb{R} \mid x > 0\}$, and let $B_r(p) = \{(\bar{x}_1, \ldots, \bar{x}_n) \in \mathbb{R}^n \mid (x_1 - \bar{x}_1)^2 + \ldots + (x_n - \bar{x}_n)^2 < r^2\} \subseteq \mathbb{R}^n$ be the set (or *ball*, *neighborhood*, *disk*) around point p with radius r. We specify the subsets of $\mathbb{R}^n$ that we consider open sets and put into our topology, called T_n^u, as follows: $T_n^u = \{V \subseteq \mathbb{R}^n \mid \forall p \in V \; \exists r \in \mathbb{R}^+ : B_r(p) \subseteq V\}$. T_n^u is indeed a topology on $\mathbb{R}^n$ since it fulfills the conditions of a general topology as we show now. We look at the first condition $\emptyset \in T_n^u$. If $V \in T_n^u$, every point p of V is an element of a ball $B_r(p)$ contained entirely within V. That is, there are no points in V for which one can only find a ball some of whose elements are not in V. As there are no elements in $\emptyset$, the condition is vacuously true. $\mathbb{R}^n \in T_n^u$ holds since all balls $B_r(p)$ are a subset of $\mathbb{R}^n$ and one can find for each point $p \in \mathbb{R}^n$ a ball $B_r(p)$ that is entirely contained in $\mathbb{R}^n$. For the second condition, we assume that $V_1, V_2 \in T_n^u$. We show that then also $V_1 \cap V_2 \in T_n^u$ holds. Let $p \in V_1 \cap V_2$. This means that $p \in V_1$ and $p \in V_2$. Since V_1 and V_2 are open sets, there must be radii $r_1, r_2 \in \mathbb{R}^+$ and balls $B_{r_1}(p) \subseteq V_1$ and $B_{r_2}(p) \subseteq V_2$. W.l.o.g., let $r_1 \leq r_2$. Then $B_{r_1}(p) \subseteq B_{r_2}(p)$ and consequently $B_{r_1}(p) \subseteq V_2$. Since we already know that $B_{r_1}(p) \subseteq V_1$, we obtain that $B_{r_1}(p) \subseteq V_1 \cap V_2$. This means that $V_1 \cap V_2 \in T_n^u$. Finally, to show the third condition, let $S \subseteq T_n^u$, that is, S is a set of open sets of T_n^u. We show that $\bigcup_{V \in S} V \in T_n^u$. Let $p \in \bigcup_{V \in S} V$. Then $\exists W \in S : p \in W$. Since W is an open set, there exists a radius $r \in \mathbb{R}^+$ such



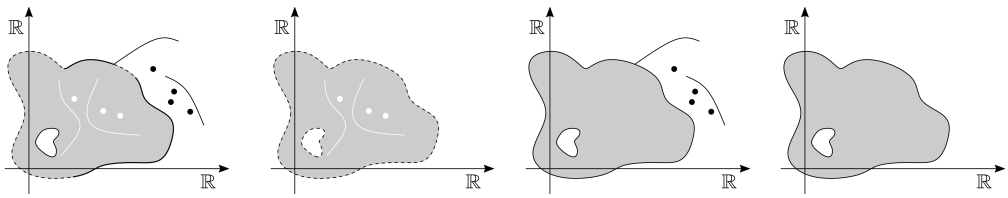
(a) a ball $B_r(p)$ around a point p with radius r (b) an open set (c) a closed set

Fig. 3. Open and closed sets in $(\mathbb{R}^2, T_2^u)$. A straight (dashed) boundary line indicates that points on the line are (are not) elements of the set enclosed by the line.



(a) a crisp region object V (b) the interior $int_{T_2^u}(V)$ (c) the closure $cl_{T_2^u}(V)$ (d) the boundary $bd_{T_2^u}(V)$

Fig. 4. Interior, closure, and boundary of a crisp region object in $(\mathbb{R}^2, T_2^u)$.

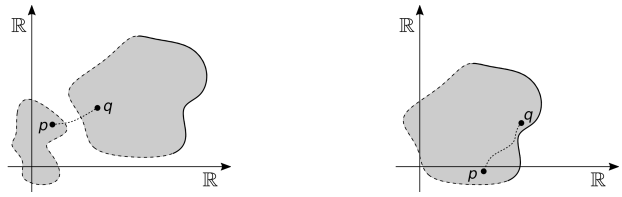


(a) a set W with irregularities (b) the interior $int_{T_2^u}(W)$ (c) the closure $cl_{T_2^u}(W)$ (d) the regular closure $cl_{T_2^u}(int_{T_2^u}(W))$

Fig. 5. Effect of the interior, closure, and regular closure operators on a set with geometric anomalies in $(\mathbb{R}^2, T_2^u)$.

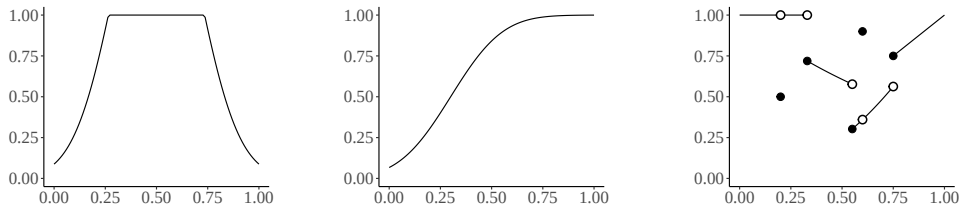
that $B_r(p) \subseteq W \subseteq W \cup \bigcup_{V \in S - \{W\}} V = \bigcup_{V \in S} V$. This means that $\bigcup_{V \in S} V \in T_n^u$. The family of all closed sets in $(\mathbb{R}^n, T_n^u)$ is $T_n^{u'} = \{\mathbb{R}^n - V \mid V \in T_n^u\}$. T_n^u is called the *usual* (or *standard*) *topology* on $\mathbb{R}^n$. Figure 3 provides the geometric interpretation and illustration of a ball, an open set, and a closed set in $(\mathbb{R}^2, T_2^u)$.

The topological concepts of *interior*, *closure*, and *boundary* in $(\mathbb{R}^n, T_n^u)$ are applications of the corresponding general concepts defined in Section 3.2.3. Figure 4 illustrates these concepts for a crisp region object in $(\mathbb{R}^2, T_2^u)$. General open and closed sets can be afflicted with anomalies. For instance, in $(\mathbb{R}, T_1^u)$ these anomalies can be isolated points. In $(\mathbb{R}^2, T_2^u)$ these anomalies can be isolated points, dangling lines, punctures, and cuts (see Figure 5(a)). Figures 5(b) and 5(c) show which anomalies can be repaired by the *interior* operator and the *closure* operator respectively defined in Section 3.2.3. All anomalies can be removed by *regular closure* as Figure 5(d) shows. Based on the definitions in Section 3.2.3, the geometric interpretation of a *connected* set in $(\mathbb{R}^n, T_n^u)$ is that for any two points in the set there is a path that connects both points and is located completely in the set; otherwise, the set is *disconnected*. Figure 6 provides examples for connected and disconnected sets in $(\mathbb{R}^2, T_2^u)$.



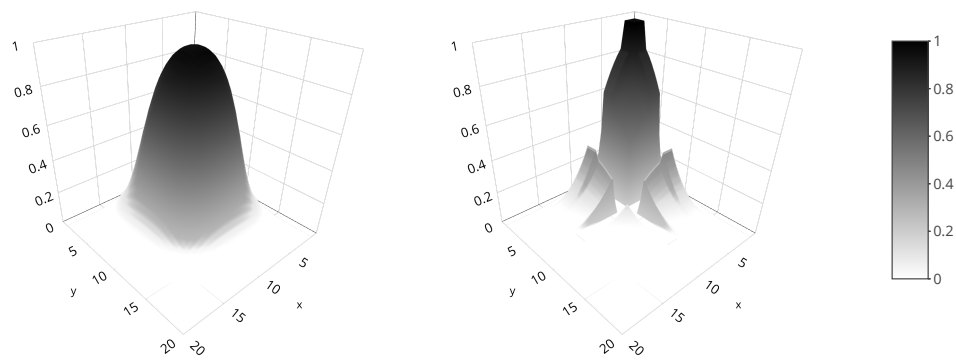
(a) a disconnected set (b) a connected set

Fig. 6. Connected and disconnected sets in $(\mathbb{R}^2, T_2^u)$.



(a) a continuous membership function (b) a continuous membership function (c) a (neither upper nor lower) semi-continuous membership function

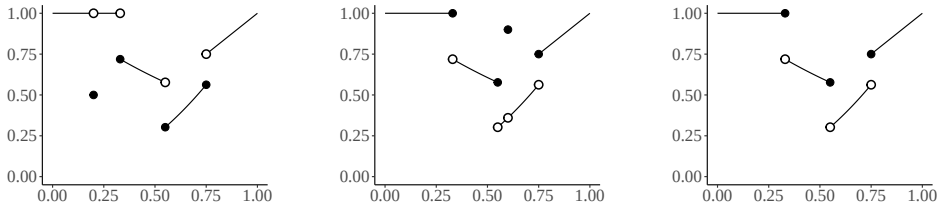
Fig. 7. Examples of membership functions $\mu_{\tilde{A}} : \mathbb{R} \rightarrow [0, 1]$.



(a) a continuous membership function (b) an upper semi-continuous membership function

Fig. 8. Examples of membership functions $\mu_{\tilde{A}} : \mathbb{R}^2 \rightarrow [0, 1]$. The gray-scale bar, which is employed in the remainder of this article, depicts how the membership degrees are associated with the colors.

The usual topology $(\mathbb{R}^n, T_n^u)$ is a general, that is, crisp, topology. Such a general topology may give rise to a fuzzy topology. To represent the fuzzy spatial nature of real-world phenomena, we deploy *continuous membership functions* as well as *lower* and *upper semi-continuous membership functions* $\mu_{\tilde{A}} : X \rightarrow [0, 1]$. Intuitively, *continuous membership functions* allow us to model smooth and seamless transitions between membership degrees. *Semi-continuous membership functions* (see Section 3.2.4), in addition, permit a finite number of discontinuities. Figures 7 and 8 depict examples of continuous and semi-continuous membership functions with $X = \mathbb{R}$ and $X = \mathbb{R}^2$, respectively. In Figure 7(c), filled and empty circles are used to represent the actual and missing values of the function at discontinuity points. Figure 8(b) shows discontinuities drawn as "vertical walls" in the graph to depict abrupt changes in the transition between the membership degrees of nearby points.



(a) the fuzzy interior (b) the fuzzy closure (c) the fuzzy regular closure

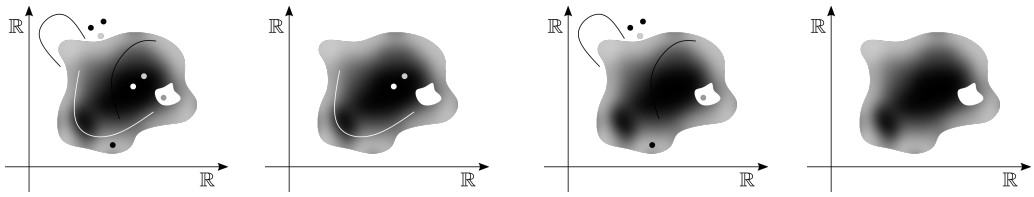
Fig. 9. Effect of applying the operations of fuzzy interior, fuzzy closure, and fuzzy regular closure on the semi-continuous membership function shown in Figure 7(c).

The family $\mathcal{T}_n^u$ of lower semi-continuous functions in $(\mathbb{R}^n, T_n^u)$ forms a fuzzy topology, which is called the *induced fuzzy topology* from the usual topology T_n^u on $\mathbb{R}^n$ [84]. The fuzzy sets with a lower semi-continuous membership function are the open sets in $\mathcal{T}_n^u$, and those with an upper semi-continuous membership function are the closed sets. The fuzzy sets with continuous membership functions are open and closed, and called *clopen*. We use the fuzzy topological space $(\mathbb{R}^n, \mathcal{T}_n^u)$ as the basis of our definition of fuzzy spatial data types in Sections 3.5 to 3.7. In particular, the definition of fuzzy lines makes use of the fuzzy topological space $(\mathbb{R}, \mathcal{T}_1^u)$ while the definition of fuzzy regions deploys the fuzzy topological space $(\mathbb{R}^2, \mathcal{T}_2^u)$.

Lower and upper semi-continuity functions allow *isolated* discontinuities. This leads to abrupt changes of membership degrees at single points or abrupt changes on both sides of linear parts. As a consequence, this causes geometric anomalies in fuzzy line and fuzzy region objects. Examples are isolated points, dangling lines, cuts, and punctures. As previously discussed, the regular closure repairs these anomalies. In the case of the fuzzy topological space $(\mathbb{R}^n, \mathcal{T}_n^u)$, we apply the concept of *fuzzy regular closure*, where the closure operator and the interior operator deal with discontinuities differently. Let $\tilde{A}, \tilde{B}$ be fuzzy sets in the fuzzy topological space $(\mathbb{R}^n, \mathcal{T}_n^u)$. The fuzzy closure operator $cl_{\mathcal{T}_n^u}$ on X (in our case $\mathbb{R}^2$) is a mapping function $[0,1]^X \rightarrow [0,1]^X$ that satisfies the Kuratowski closure axioms [58]: (i) $cl_{\mathcal{T}_n^u}(\emptyset) = \emptyset$, (ii) $\tilde{A} \subseteq cl_{\mathcal{T}_n^u}(\tilde{A})$, (iii) $cl_{\mathcal{T}_n^u}(cl_{\mathcal{T}_n^u}(\tilde{A})) = cl_{\mathcal{T}_n^u}(\tilde{A})$, and (iv) $cl_{\mathcal{T}_n^u}(\tilde{A} \cup \tilde{B}) = cl_{\mathcal{T}_n^u}(\tilde{A}) \cup cl_{\mathcal{T}_n^u}(\tilde{B})$. The fuzzy interior operator $int_{\mathcal{T}_n^u}$ on X (in our case $\mathbb{R}^2$) is a mapping function $[0,1]^X \rightarrow [0,1]^X$ that satisfies the Kuratowski interior axioms: (i) $int_{\mathcal{T}_n^u}(X) = X$, (ii) $int_{\mathcal{T}_n^u}(\tilde{A}) \subseteq \tilde{A}$, (iii) $int_{\mathcal{T}_n^u}(int_{\mathcal{T}_n^u}(\tilde{A})) = int_{\mathcal{T}_n^u}(\tilde{A})$, and (iv) $int_{\mathcal{T}_n^u}(\tilde{A} \cap \tilde{B}) = int_{\mathcal{T}_n^u}(\tilde{A}) \cap int_{\mathcal{T}_n^u}(\tilde{B})$. Both operators are dual to each other.

The effect of the fuzzy interior, fuzzy closure, and fuzzy regular closure operators on fuzzy sets in the fuzzy topological space $(\mathbb{R}^n, \mathcal{T}_n^u)$ is as follows. The fuzzy interior operator sets the membership degree of discontinuity points to the lower value (if the discontinuity point does not already have the lower value). This ensures that the fuzzy set contains its interior (axiom (ii) of the fuzzy interior operator). Figures 9(a) and 10(b) depict the effect of the fuzzy interior operator on the fuzzy point sets of Figures 7(c) and 10(a), respectively. In Figure 10(b), dangling points and lines are eliminated since they symbolize abrupt changes in surrounding areal parts with membership degrees equal to 0. This operator also eliminates cuts and punctures whose points have a larger membership degree than the points in their neighborhood. Intuitively, the fuzzy closure operator sets the membership degrees of points with abrupt changes (that is, discontinuities) to the higher values. This ensures that the fuzzy set is contained in its closure (axiom (ii) of the fuzzy closure operator). Figures 9(b) and 10(c) depict the effect of the fuzzy closure on the fuzzy point sets of Figures 7(c) and 10(a), respectively. In Figure 10(c), cuts and punctures that have membership degree less than their neighborhood are eliminated.

When applying the fuzzy regular closure operator to $\tilde{A}$, that is, $cl_{\mathcal{T}_n^u}(int_{\mathcal{T}_n^u}(\tilde{A}))$, the result is a fuzzy point set without isolated discontinuities and with stronger properties than an upper



(a) a fuzzy point set $\tilde{W}$ with irregularities (b) the fuzzy interior $int_{\mathcal{T}_2^u}(\tilde{W})$ (c) the fuzzy closure $cl_{\mathcal{T}_2^u}(\tilde{W})$ (d) the fuzzy regular closure $cl_{\mathcal{T}_2^u}(int_{\mathcal{T}_2^u}(\tilde{W}))$

Fig. 10. Effect of applying the operations of fuzzy interior, fuzzy closure, and fuzzy regular closure on a fuzzy point set with geometric anomalies in $(\mathbb{R}^2, \mathcal{T}_2^u)$.

semi-continuous function. In summary, it only allows jump discontinuities along lines so that each side of a line has a continuous distribution of membership degrees, as shown in Figures 9(c) and 10(d).

3.4 Running Example

Throughout the rest of this article, we make use of a hypothetical agricultural and ecological application as a running example to illustrate the applicability of FUSA. The main goal of this application is to aid in decision making for crop area planning by managing vague geographic phenomena that are stored as fuzzy spatial objects and are manipulated by fuzzy spatial operations. This application includes synthetic examples that are based on typical fuzzy phenomena studied in research papers about real agricultural and ecological applications. From an ecological perspective, we cite polluted rivers [34], animal routes [39], and habitats of species [51]. From an agricultural perspective, we mention infected trees [31], effectiveness of insect traps [23], and air pollution [50].

In the following sections, we build up and employ the running example to exemplify the conceptual definitions of fuzzy spatial data types (Sections 3.5 to 3.7) and their operations (Section 4). For this, we use the concept of *abstract data types* (*ADT*) [73] available in most *extensible* database systems (e.g., DB2, Oracle, PostgreSQL) to embed the fuzzy spatial data types as attribute data types in relational table schemas. This means that a fuzzy spatial data type can be deployed in the same manner as a standard data type (e.g., *int*, *date*) of conventional database systems. In this sense, a fuzzy spatial data type is a *complex data type* whose internal complexity is deliberately hidden from the user and whose values (called *complex objects*) are only accessible by operations provided by the interface of the ADT. This approach has two main advantages. First, the representation of a complex object is not scattered over a set of relational tables but concentrated in the attribute value representing the complex fuzzy spatial object. Second, the implementation of a fuzzy spatial data type can be exchanged and improved without any consequences for the query language and application programs.

3.5 Fuzzy Points

In this section, we design a concept of a fuzzy point object and introduce the fuzzy spatial data type *fpoint* for this class of objects. Section 3.5.1 gives a general description of fuzzy points. Section 3.5.2 provides the formal definition of type *fpoint*. Finally, Section 3.5.3 exemplifies the applicability of fuzzy points by using our running example.

3.5.1 What are Fuzzy Points? Points are the simplest geometric abstraction, that is, the atoms, of the Euclidean space $\mathbb{R}^2$. Each element of a *crisp point set* A of $\mathbb{R}^2$ belongs definitely and totally to this set. Each element of a *fuzzy point set* $\tilde{A}$ of $\mathbb{R}^2$ only has a certain partial membership in this set. It can also partially belong to another fuzzy point set, that is, multiple set membership is possible.

For example, if we consider an air-polluted area (which is a fuzzy region), then each point of this area is fuzzy since the concentration of air pollution at a point is not necessarily 100 percent. That is, the degree of membership of such a fuzzy point in the air-polluted area is less than or equal to 1 and larger than 0. A *fuzzy point object* can represent several geographic locations, each of which is assigned some degree of membership. Fuzzy point objects can, for example, arise as the result of calculating the common points of two fuzzy line objects.

3.5.2 Formal Definition of a Fuzzy Spatial Data Type for Fuzzy Points. Formally, a fuzzy point object $\tilde{P}$ is composed of possibly many *simple fuzzy points*. A simple fuzzy point $\tilde{p}$ at (a, b) in $\mathbb{R}^2$, written $\tilde{p}(a, b)$, is a fuzzy singleton in $\mathbb{R}^2$ defined by the membership function

$$\mu_{\tilde{p}(a,b)}(x, y) = \begin{cases} m & \text{if } (x, y) = (a, b) \\ 0 & \text{otherwise} \end{cases}$$

with $m \in]0, 1]$. If $m = 1$ holds, we call $\tilde{p}(a, b)$ a *simple crisp point*. Figure 2(a) depicts an example of a simple fuzzy point. Let P_f be the set of all simple fuzzy points, and let $\tilde{p}(a, b), \tilde{q}(c, d) \in P_f$ with $a, b, c, d \in \mathbb{R}$. We call $\tilde{p}(a, b)$ and $\tilde{q}(c, d)$ *disjoint* if $(a, b) \neq (c, d)$ holds. We now define the fuzzy spatial data type *fpoint* as follows:

$$\mathit{fpoint} = \{Q \subseteq P_f \mid \forall\, \tilde{p}(a, b), \tilde{q}(c, d) \in Q : \tilde{p}(a, b) \text{ and } \tilde{q}(c, d) \text{ are } \mathit{disjoint} \ \wedge\ Q \text{ is finite}\}$$

Disjointedness of the simple fuzzy points of a fuzzy point object is required since the membership degree of each simple fuzzy point must be unique. The definition includes the *empty fpoint* object ($Q = \tilde{\emptyset}$). It is needed to represent, for example, that the fuzzy intersection of two *fpoint* objects yields the empty *fpoint* object. Figure 2(d) shows an example of an *fpoint* object composed of seven simple fuzzy points. A fuzzy point object is modeled as a multi-component object in order to ensure closure under fuzzy set operations such as *funion*, *fintersection*, and *fdifference* defined in Section 4.3.1, that is, for any two fuzzy point objects as input, these operations yield a fuzzy point object as their result. Let P_c be the set of all simple crisp points. If we replace the set P_f in the definition of type *fpoint* by the set P_c, we obtain the type *point* of crisp point objects as a proper sub-type of type *fpoint*. This means that $\tilde{p} \in \mathit{fpoint}$ implies that $p = \mathit{supp}(\tilde{p}) \in \mathit{point}$.

3.5.3 Application to Running Example. In our running example we represent two phenomena by fuzzy points: infected trees and the effectiveness of insect traps. Each infected tree is fuzzy because of the infection level of the tree. For instance, a tree can be annotated with a degree of infection by using a real value in $]0, 1]$. Note that healthy trees would have membership degree 0 and thus, they do not belong to the set of infected trees. Each insect trap is also fuzzy because of its effectiveness in capturing insects and plagues at a particular location. Such effectiveness can be measured, for example, based on the ratio of the number of captured insects and the maximum capacity of the trap for capturing insects. Multiple set membership is possible in this scenario. For instance, a point can have membership degree 0.8 to indicate the level of infection of a tree and at the same time membership degree 0.5 to indicate the effectiveness level of a trap.

By embedding the fuzzy spatial data type *fpoint* as an ADT (Section 3.4) in spatial database systems, we can store infected trees and the effectiveness of insect traps in the relational table schemas *itree* and *trap*, respectively. Both schemas contain attributes *inf_tree* and *tp_loc* respectively representing the locations of infected trees and insect traps as fuzzy point objects. We specify these two schemas as follows.

```
itree(id : int, type : varchar(50), level : varchar(30), inf_tree : fpoint)
trap(id : int, when : timestamp, type : varchar(50), tp_loc : fpoint)
```

3.6 Fuzzy Lines

In this section, we focus on designing fuzzy line objects by introducing the fuzzy spatial data type *fline*. A conceptual description is given in Section 3.6.1. Section 3.6.2 introduces the formal definition of type *fline*. Finally, Section 3.6.3 employs fuzzy line objects to represent fuzzy linear geographical phenomena in our running example.

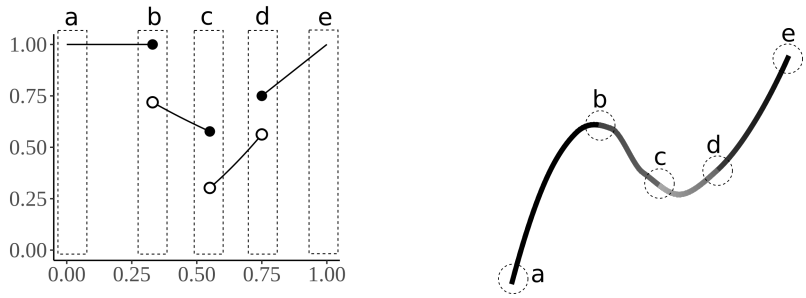
3.6.1 What are Fuzzy Lines? Lines are the one-dimensional geometric abstraction of linear features in space. A crisp line object represents a particular point set of the Euclidean space $\mathbb{R}^2$ and corresponds to a finite collection of non-self-intersecting simple lines (curves or segments). This means that each pair of simple lines is either disjoint or meets in a common end point. Each point of a simple line is crisp and definitely and totally belongs to the crisp line object. The connected components of a crisp line object are called *blocks*; any two blocks are disjoint. Formal definitions of crisp line objects are given in Schneider [67] and Schneider and Behr [71].

The fuzzy domain extends the notion of crisp line objects by allowing partial membership of points in simple lines. A fuzzy line object has the same linear geometry as a crisp line object. But, in addition, each of its points is associated with a membership degree indicating to which extent a point belongs to the line object. A membership function is required to model the smooth change of membership degrees, possibly with a finite number of abrupt interruptions. For example, the pollutant concentration of a river can be represented by the line geometry of the river where each point represents the degree or concentration of pollution at that location. The concentration is larger than 0 for all points of the fuzzy line but usually different at different locations.

3.6.2 Formal Definition of a Fuzzy Spatial Data Type for Fuzzy Lines. We assume a total order on $\mathbb{R}^2$ that is given by the lexicographic order "$<$" on the coordinates (first x, then y) of the points of $\mathbb{R}^2$. Let $\tilde{E}$ be a fuzzy set defined by a membership function $\mu_{\tilde{E}} : [0,1] \rightarrow [0,1]$. Formally, a *simple fuzzy line* $\tilde{l}$ is defined by a *constructor function* $f_{\tilde{l}} : [0,1] \rightarrow \mathbb{R}^2$ and a *membership function* $\mu_{\tilde{l}} : \mathbb{R}^2 \rightarrow [0,1]$ such that

(i) $f_{\tilde{l}}$ is continuous
(ii) $\mu_{\tilde{l}}$ is either continuous or finitely upper semi-continuous
(iii) $\tilde{E} = cl_{\mathcal{T}_1^u}(int_{\mathcal{T}_1^u}(\tilde{E}))$
(iv) $\forall\, a \in [0,1] : \mu_{\tilde{l}}(f_{\tilde{l}}(a)) = \mu_{\tilde{E}}(a)$, i.e., $\mu_{\tilde{E}} = \mu_{\tilde{l}} \circ f_{\tilde{l}}$
(v) $\tilde{l}$ is bounded
(vi) $\forall\, a, b \in]0,1[: a \neq b \Rightarrow f_{\tilde{l}}(a) \neq f_{\tilde{l}}(b)$
(vii) $\forall\, a \in \{0,1\}\ \forall\, b \in]0,1[: f_{\tilde{l}}(a) \neq f_{\tilde{l}}(b)$
(viii) $f_{\tilde{l}}(0) < f_{\tilde{l}}(1) \vee (f_{\tilde{l}}(0) = f_{\tilde{l}}(1) \wedge \forall\, a \in]0,1[: f_{\tilde{l}}(0) < f_{\tilde{l}}(a))$
(ix) $\forall\, p \in \mathbb{R}^2 - f_{\tilde{l}}([0,1]) : \mu_{\tilde{l}}(p) = 0$

Conditions (i) and (ii) determine the structure of a simple fuzzy line. Function $f_{\tilde{l}}$ alone models a continuous simple *crisp* curve. The points at the *positions* 0 and 1 of $f_{\tilde{l}}$, that is, $f_{\tilde{l}}(0)$ and $f_{\tilde{l}}(1)$, are the *end points* or *boundary points* of $\tilde{l}$; the points of $f_{\tilde{l}}(]0,1[)$ are the *interior points* of $\tilde{l}$. Since the interval $[0,1]$ in $f_{\tilde{l}}$ is connected and closed and since $f_{\tilde{l}}$ is a continuous mapping, the image $f_{\tilde{l}}(]0,1[)$ (and thus $\tilde{l}$) is connected and closed too. Function $\mu_{\tilde{l}}$ employs the left open interval $]0,1]$ for assigning membership degrees greater than 0 to the points of $f_{\tilde{l}}$ only. This avoids the contradiction of including points with membership degree equal to 0, that is, points that actually do not belong to the line. Further, $\mu_{\tilde{l}}$ must be continuous or finitely semi-continuous (Section 3.2.4). This requirement enables the modeling of smooth transitions of membership degrees but also permits finitely many points of discontinuity since interruptions can appear in geospatial applications. If in a simple



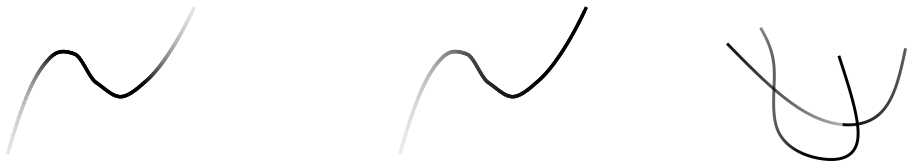
(a) a fuzzy set $\tilde{E}$ (b) a fuzzy line object $\tilde{l}$

Fig. 11. An illustration of how $\tilde{E}$ maps the position of a point $p \in \mathbb{R}^2$ in a fuzzy line object $\tilde{l}$. The letters indicate the positions of the ends points (a and e) and discontinuity points (b to d) of $\tilde{l}$. For instance, this means that $\mu_{\tilde{E}}(b) = \mu_{\tilde{l}}(f_{\tilde{l}}(b))$, where b refers to a particular position (0.33) of a point in $\tilde{l}$.

fuzzy line $\tilde{l}$ the membership function $\mu_{\tilde{l}}$ maps all its points to the membership degree 1 as a special case, $\tilde{l}$ is called a *simple crisp line*. Conditions (iii) and (iv) eliminate *isolated discontinuities* in the transitions of the membership degrees that belong to the points in $\tilde{l}$. We are unable to *directly* apply the concept of fuzzy regularization on linear objects in $\mathbb{R}^2$ to eliminate such discontinuities because the topological space $(\mathbb{R}^2, \mathcal{T}_2^u)$ is defined on areal representations only. On the other hand, we can make use of a fuzzy set $\tilde{E}$ with the membership function $\mu_{\tilde{E}}$ derived from the composition of $\mu_{\tilde{l}}$ and $f_{\tilde{l}}$. This composition maps $s \in [0, 1]$ to $\mu_{\tilde{l}}(f_{\tilde{l}}(s)) \in [0, 1]$ such that s is mapped to the position of a point $p \in \mathbb{R}^2$ in the fuzzy line object $\tilde{l}$. Figure 11 depicts an example of this mapping by using the upper semi-continuous function shown in Figure 9(c). Since $\mu_{\tilde{E}}$ assigns membership degrees to the points of $\tilde{l}$ and its membership function is defined on $(\mathbb{R}, \mathcal{T}_1^u)$, we can apply the concept of fuzzy regularization by employing the closure and interior operators $cl_{\mathcal{T}_1^u}$ and $int_{\mathcal{T}_1^u}$ on fuzzy sets (as discussed in Sections 3.2.3 and 3.3) to eliminate isolated discontinuities. Hence, intuitively, the membership function of a simple fuzzy line object only allows jumps discontinuities, as shown in Figure 11. Condition (v) avoids the definition of infinite fuzzy lines by requiring the boundedness property. Condition (vi) allows loops ($f_{\tilde{l}}(0) = f_{\tilde{l}}(1)$) but prohibits the equality of interior points and thus self-intersections since they do not occur in spatial reality. Condition (vii) disallows the equality of an interior point with an end point. Condition (viii) requires that either the left end point of the simple line is smaller than its right end point, or the simple line is closed with $f_{\tilde{l}}(0)$ as its leftmost point. The main reason of the conditions (iii) to (viii) is to ensure the uniqueness of the representation of a simple fuzzy line. For example, consider a simple line that intersects itself one time and is not closed. An alternative to model this configuration is to model the intersection point explicitly with three incident simple lines from which one line forms a closed loop. This alternative concept leads to a unique representation. Condition (ix) ensures that all points not belonging to $f_{\tilde{l}}$ have the membership degree equal to 0. This completes the definition of the membership function on $\mathbb{R}^2$, which guarantees that all points in the space have a valid membership degree.

Let the application of a function f to a set X of values be defined as $f(X) = \{f(x) \mid x \in X\}$. Based on the above definition, a simple fuzzy line $\tilde{l}$ is given as the fuzzy point set $\tilde{l} = \{(p, \mu_{\tilde{l}}(p)) \mid p \in f_{\tilde{l}}([0, 1])\}$. Figures 12(a) and 12(b) depict two simple fuzzy line objects that have the same geometric shape but different membership functions.

Let SL_f be the set of all simple fuzzy lines and $\tilde{l}_1, \tilde{l}_2 \in SL_f$. We define the crisp relationships *c-disjoint* and *c-meet* as the following Boolean functions (that is, predicates):



(a) a simple fuzzy line object based on the membership function of Figure 7(a)

(b) a simple fuzzy line object based on the membership function of Figure 7(b)

(c) a complex fuzzy line object with six simple fuzzy lines

Fig. 12. Examples of fuzzy line objects.

(i) $c\text{-}disjoint(\tilde{l}_1, \tilde{l}_2) :\Leftrightarrow supp(\tilde{l}_1) \cap supp(\tilde{l}_2) = \varnothing$

(ii) $c\text{-}meet(\tilde{l}_1, \tilde{l}_2) :\Leftrightarrow f_{\tilde{l}_1}(]0,1[) \cap f_{\tilde{l}_2}(]0,1[) = \varnothing \wedge \{f_{\tilde{l}_1}(0), f_{\tilde{l}_1}(1)\} \cap f_{\tilde{l}_2}(]0,1[) = \varnothing \wedge$
$f_{\tilde{l}_1}(]0,1[) \cap \{f_{\tilde{l}_2}(0), f_{\tilde{l}_2}(1)\} = \varnothing \wedge \{f_{\tilde{l}_1}(0), f_{\tilde{l}_1}(1)\} \cap \{f_{\tilde{l}_2}(0), f_{\tilde{l}_2}(1)\} \neq \varnothing$

Let $\tilde{l}_1, \ldots, \tilde{l}_n \in SL_f$ for some $n \in \mathbb{N}$. For all $1 \leq i, j \leq n$ and for all $a, k \in \{0,1\}$ let $V^a_{\tilde{l}_i} = \{(j,k) \mid f_{\tilde{l}_i}(a) = f_{\tilde{l}_j}(k)\}$. This means that $V^a_{\tilde{l}_i}$ registers each simple fuzzy line with its respective end point that is incident to $\tilde{l}_i$. Note that always $(i,a) \in V^a_{\tilde{l}_i}$ holds. For a set X let $\mathcal{P}^{\text{fin}}(X)$ be the powerset of all finite subsets of X, that is, $\mathcal{P}^{\text{fin}}(X) = \{Y \subseteq X \mid |Y| \text{ is finite}\}$. We are now able to define the fuzzy spatial data type *fline* as follows:

$$
\begin{aligned}
fline = \{\textstyle\bigcup_{\tilde{l} \in S} \tilde{l} \mid \text{(i)} \;\; & S \in \mathcal{P}^{\text{fin}}(SL_f) \\
\text{(ii)} \;\; & \forall\, 1 \leq i < j \leq |S| : c\text{-}disjoint(\tilde{l}_i, \tilde{l}_j) \vee c\text{-}meet(\tilde{l}_i, \tilde{l}_j) \\
\text{(iii)} \;\; & \forall\, 1 \leq i \leq |S| \;\forall\, a \in \{0,1\} : (|V^a_{\tilde{l}_i}| = 1) \vee (|V^a_{\tilde{l}_i}| > 2) \\
\text{(iv)} \;\; & \forall\, 1 \leq i \leq |S| \;\forall\; a \in \{0,1\} \;\forall\; (j,k) \in V^a_{\tilde{l}_i} : \mu_{\tilde{l}_i}(f_{\tilde{l}_i}(a)) = \mu_{\tilde{l}_j}(f_{\tilde{l}_j}(k))\}
\end{aligned}
$$

Condition (i) states that a fuzzy line object consists of the fuzzy union of finitely many simple fuzzy lines. Different fuzzy line objects can include a different finite number of simple fuzzy lines. Further, this means that the membership function $\mu_{\tilde{L}}$ of $\tilde{L} \in fline$ is a finitely piecewise and finitely semi-continuous function (see Section 3.2.4). Condition (ii) requires that the supports of any two simple fuzzy lines belonging to an *fline* object are either disjoint or adjacent. Condition (iii) ensures uniqueness of representation. If $|V^a_{\tilde{l}_i}| = 2$ would be allowed, two incident simple fuzzy lines could be alternatively joined into a single one. Condition (iv) ensures consistency of representation in the sense that the common end points of simple fuzzy lines must have the same membership degree. Let SL_c be the set of all simple crisp lines. If we replace the set SL_f in the definition of type *fline* by the set SL_c, we obtain the type *line* of crisp line objects as a proper sub-type of type *fline*. This means that $\tilde{L} \in fline$ implies that $L = supp(\tilde{L}) \in line$. Figure 12(c) shows an example of an *fline* object composed of six simple fuzzy lines.

From an *fline* object we can identify (maximal) connected components called *fuzzy blocks*. Let $\tilde{L} \in fline$. Then a set $S \in \mathcal{P}^{\text{fin}}(SL_f)$ exists such that $\tilde{L} = \bigcup_{\tilde{l} \in S} \tilde{l}$ and the properties of type *fline* hold. Let $S_1, \ldots, S_m \subseteq S$ for some $m \in \mathbb{N}$. The fuzzy point sets $\tilde{B}_1 = \bigcup_{\tilde{l} \in S_1} \tilde{l}, \ldots, \tilde{B}_m = \bigcup_{\tilde{l} \in S_m} \tilde{l}$ are the connected components of $\tilde{L}$ if, and only if,

(i) $\tilde{L} = \bigcup_{i=1}^{m} \bigcup_{\tilde{l} \in S_i} \tilde{l} = \bigcup_{i=1}^{m} \tilde{B}_i$

(ii) $\forall\, 1 \leq i < j \leq m : \tilde{B}_i \cap \tilde{B}_j = \tilde{\varnothing}$

(iii) $\forall\, 1 \leq k \leq m \;\forall\, 1 \leq i < j \leq |S_k| \;\exists\, \tilde{l}^{ij}_1, \ldots, \tilde{l}^{ij}_{n_{ij}} \in S_k :$
$c\text{-}meet(\tilde{l}_i, \tilde{l}^{ij}_1) \wedge c\text{-}meet(\tilde{l}^{ij}_1, \tilde{l}^{ij}_2) \wedge \ldots \wedge c\text{-}meet(\tilde{l}^{ij}_{n_{ij}-1}, \tilde{l}^{ij}_{n_{ij}}) \wedge c\text{-}meet(\tilde{l}^{ij}_{n_{ij}}, \tilde{l}_j)$

Conditions (i) and (ii) partition $\tilde{L}$ into a collection of disjoint fuzzy point sets, each of which is supposed to be a fuzzy block. That each such fuzzy point set is indeed a fuzzy block is ensured by Condition (iii) that defines the connectivity property within each fuzzy block. Figure 12(c) shows an example of an *fline* object composed of six fuzzy blocks.

3.6.3 Application to Running Example. In our running example, we represent two phenomena by fuzzy line objects: polluted rivers and animal routes. Each polluted river is fuzzy because of different concentrations of pollution along the course of the river. For instance, a membership function modeling a polluted river can be defined according to the pollution level of a chemical product at particular locations. Each animal route is also fuzzy because of the vagueness of determining the presence of an animal in a specific point of the route. For instance, the chance to find an animal in a point of its route can be modeled according to collected data from remote sensing devices. Fuzzy line objects representing different types of geographical phenomena can intersect in common points, thus allowing these points to have multiple set membership.

By embedding the fuzzy spatial data type *fline* as an ADT (Section 3.4) in spatial database systems, we can store polluted rivers and animal routes in the relational table schemas *polluted_river* and *animal,* respectively. Both schemas contain the attributes *course* and *animal_route* respectively representing the course of polluted rivers and the animal routes as fuzzy line objects. We specify these two tables as follows:

```
polluted_river(id : int, name : varchar(50), course : fline)
animal(id : int, species : varchar(50), animal_route : fline)
```

3.7 Fuzzy Regions

In this section, we focus on designing fuzzy region objects by introducing the fuzzy spatial data type *fregion.* We provide a conceptual description in Section 3.7.1 and introduce the formal definition of type *fregion* in Section 3.7.2. Finally, Section 3.7.3 employs fuzzy region objects to represent fuzzy areal geographical phenomena in our running example.

3.7.1 What are Fuzzy Regions? Regions are the two-dimensional geometric abstraction for areal components in space. A crisp region object corresponds to a finite collection of simple regions called *faces.* Each face may have disjoint holes and satisfies certain set-theoretic and topological properties such as boundedness and connectedness. Each point of a face is crisp and definitely and totally belongs to the crisp region object. Further, the faces of a crisp region object are disjoint or adjacent. Formal definitions of crisp region objects are given in Schneider [67] and Schneider and Behr [71].

The fuzzy domain extends the notion of crisp region objects by allowing partial membership in the points of the simple regions (that is, faces). A fuzzy region object has the same areal geometry as a crisp region object. But, in addition, it may have a vague boundary and/or a vague interior. That is, each of its points is associated with a membership degree indicating to which extent a point belongs to the region object. A membership function is supposed to model a smooth change of membership degrees, possibly with a finite number of interruptions. There are at least two reasons that prevent a definition of a fuzzy region simply as an arbitrary fuzzy point set. The first reason is the necessary application of regularization to avoid geometric anomalies like isolated or dangling line or point features and missing lines and points in the form of cuts and punctures. This process leads to the nice property that a fuzzy region object is closed under (appropriately defined) geometric operations (see Section 4). The second reason is that applications may exhibit a finite number of discontinuity gaps in a fuzzy region object.

(a) a simple fuzzy region object based on the membership function of Figure 8(a)

(b) a simple fuzzy region object

(c) a complex fuzzy region object with three simple fuzzy regions

Fig. 13. Examples of fuzzy region objects.

3.7.2 Formal Definition of a Fuzzy Spatial Data Type for Fuzzy Regions. The formal foundation of defining a *simple fuzzy region* $\tilde{r}$, which we will also call a *fuzzy face*, consists in the fuzzy topologically space $(\mathbb{R}^2, \mathcal{T}_2^u)$ (Section 3.3) and in a membership function $\mu_{\tilde{r}} : \mathbb{R}^2 \rightarrow [0, 1]$. However, using arbitrary fuzzy sets may lead to geometric anomalies like isolated or dangling line and point features as well as missing lines and points in the form of cuts and punctures [71]. To correct this, we apply the concept of *fuzzy regularization* by deploying the *closure* and *interior* operators on fuzzy sets (formally defined in Sections 3.2.3 and 3.3). Let $int_{\mathcal{T}_2^u}$ and $cl_{\mathcal{T}_2^u}$ be the *interior* and *closure* operators respectively regarding $(\mathbb{R}^2, \mathcal{T}_2^u)$. Then $\tilde{r}$ is defined by a membership function $\mu_{\tilde{r}} : \mathbb{R}^2 \rightarrow [0, 1]$ such that

(i) $\tilde{r} = cl_{\mathcal{T}_2^u}(int_{\mathcal{T}_2^u}(\tilde{r}))$
(ii) $\tilde{r}$ is bounded and connected
(iii) $\mu_{\tilde{r}}$ is either continuous or finitely upper semi-continuous
(iv) $frontier(\tilde{r}) \in fline$

Condition (i) defines that a simple region is a *regular closed fuzzy set*. It ensures that the interior of $\tilde{r}$ is free of geometric anomalies, including the discontinuities introduced by its semi-continuous membership function (see condition (iii)). Recall that $cl_{\mathcal{T}_2^u}$ removes cuts and punctures by appropriately adding points, while $int_{\mathcal{T}_2^u}$ removes dangling point and lines features since they are of lower dimension. Furthermore, $cl_{\mathcal{T}_2^u}$ introduces a fuzzy boundary (similar to a crisp boundary in the ordinary point-set topological sense) separating the points of a closed set from its exterior. Condition (ii) specifies boundedness and connectedness constraints on simple fuzzy regions as they apply to real-world fuzzy spatial objects. Condition (iii) requires a smooth distribution of membership degrees with possible exceptions. If the membership function is piecewise, the fuzzy region object may include a finite number of parts called holes that do not belong to it. Condition (iv) leverages the notion of fuzzy frontier of a fuzzy set (Section 3.2.3) to characterize the frontier of a simple fuzzy region as either completely fuzzy, or completely crisp, or partially crisp and fuzzy. A completely fuzzy frontier of a fuzzy region object only contains points with membership degree in $]0, 1[$; a completely crisp frontier only consists of points with membership degree equal to 1; otherwise, the frontier is partially crisp and fuzzy. Especially because of the last kind of possible frontier, this condition ensures that the frontier of $\tilde{r}$ is not arbitrary but a fuzzy line object. This prohibits parts that are simple fuzzy points. Figures 13(a) and 13(b) show examples of simple fuzzy region objects.

The combination of all conditions ensures that the support of a simple fuzzy region results in a valid *simple crisp region* (that is, satisfies the same properties in a crisp topological space). This means that $\tilde{r}$ in $(\mathbb{R}^2, \mathcal{T}_2^u)$ and $supp(\tilde{r})$ in $(\mathbb{R}^2, T_2^u)$ satisfy the same set-theoretic and topological properties.

Let SR_f be the set of all simple fuzzy regions, and let $\tilde{r}_1, \tilde{r}_2 \in SR_f$. We overload the crisp Boolean functions (that is, predicates) *c-disjoint* and *c-meet* as:

$$
\begin{aligned}
&\text{(i)}\ \ \textit{c-disjoint}(\tilde{r}_1, \tilde{r}_2) :\Leftrightarrow \textit{supp}(\tilde{r}_1) \cap \textit{supp}(\tilde{r}_2) = \emptyset \\
&\text{(ii)}\ \ \textit{c-meet}(\tilde{r}_1, \tilde{r}_2) \quad :\Leftrightarrow \textit{int}_{T_2^u}(\textit{supp}(\tilde{r}_1)) \cap \textit{int}_{T_2^u}(\textit{supp}(\tilde{r}_2)) = \emptyset \wedge \\
&\qquad\qquad\qquad\qquad\qquad\qquad \textit{bd}_{T_2^u}(\textit{supp}(\tilde{r}_1)) \cap \textit{bd}_{T_2^u}(\textit{supp}(\tilde{r}_2)) \neq \emptyset \wedge \\
&\qquad\qquad\qquad\qquad\qquad\qquad \textit{bd}_{T_2^u}(\textit{supp}(\tilde{r}_1)) \cap \textit{bd}_{T_2^u}(\textit{supp}(\tilde{r}_2)) \in \textit{point} \wedge \\
&\qquad\qquad\qquad\qquad\qquad\qquad \textit{bd}_{T_2^u}(\textit{supp}(\tilde{r}_1)) \cap \textit{int}_{T_2^u}(\textit{supp}(\tilde{r}_2)) = \emptyset \wedge \\
&\qquad\qquad\qquad\qquad\qquad\qquad \textit{int}_{T_2^u}(\textit{supp}(\tilde{r}_1)) \cap \textit{bd}_{T_2^u}(\textit{supp}(\tilde{r}_2)) = \emptyset
\end{aligned}
$$

The predicate *c-meet* yields true if the supports of two fuzzy regions meet in single points. We are now able to define the fuzzy spatial data type *fregion* as follows:

$$
\begin{aligned}
\textit{fregion} = \{\textstyle\bigcup_{\tilde{r} \in S} \tilde{r} \mid &\text{(i)}\ \ S \in \mathcal{P}^{\text{fin}}(SR_f) \\
&\text{(ii)}\ \ \forall\, 1 \leq i < j \leq |S| : \textit{c-disjoint}(\tilde{r}_i, \tilde{r}_j) \vee \textit{c-meet}(\tilde{r}_i, \tilde{r}_j) \\
&\text{(iii)}\ \forall\, 1 \leq i < j \leq |S|\ \forall\, (x, y) \in \textit{bd}_{T_2^u}(\textit{supp}(\tilde{r}_i)) \cap \textit{bd}_{T_2^u}(\textit{supp}(\tilde{r}_j)) : \\
&\qquad \mu_{\tilde{r}_i}(x, y) = \mu_{\tilde{r}_j}(x, y)\}
\end{aligned}
$$

Condition (i) states that a fuzzy region object is the fuzzy union of finitely many simple fuzzy regions. The use of the powerset of all finite subsets of SR_f permits that different *fregion* objects may have a different finite number of simple fuzzy regions. Further, this means that the membership function $\mu_{\tilde{r}}$ of $\tilde{r} \in \textit{fregion}$ is a finitely piecewise and finitely semi-continuous function (see Section 3.2.4). Condition (ii) requires that the supports of any two simple fuzzy regions belonging to an *fregion* object are either disjoint or adjacent. Condition (iii) ensures consistency of representation by guaranteeing that the boundary points of adjacent simple fuzzy regions have the same membership degree. Let SR_c be the set of all simple crisp regions. If we replace the set SR_f in the definition of type *fregion* by the set SR_c, we obtain the type *region* of crisp region objects as a proper sub-type of type *fregion*. This means that $\tilde{R} \in \textit{fregion}$ implies that $R = \textit{supp}(\tilde{R}) \in \textit{region}$. Figure 13(c) shows an example of a complex fuzzy region object composed of three simple fuzzy region objects.

3.7.3 Application to Running Example. In our running example, we introduce two phenomena represented by fuzzy region objects: habitats of species and air pollution. Each habitat is fuzzy because of the vagueness of determining the presence of a species in a specific location of its habitat. For instance, a membership function can model the presence of a species in a given point by means of collected data from remote sensing devices. Each area representing air pollution is also fuzzy because of the different percentages of concentrations of pollutants (e.g., NO_2). For example, a method could estimate the spatial distribution and the concentration levels of NO_2 air pollution. Another method could model a transition zone between two concentrations levels. For instance, in case of a fuzzy region object representing the level *moderate* and another fuzzy region object referring to the class *high*, its intersecting area could model the fuzzy transition zone between moderate and high; thus, multiple set membership is possible.

By embedding the fuzzy spatial data type *fregion* as an ADT (Section 3.4) in spatial database systems, we can store habitats of species (attribute *habitat*) and air pollution areas (attribute *area*) as fuzzy spatial objects in the relational table schemas *habitat* and *air_pollution,* respectively. We obtain the following two table schemas:

```
habitat(id : int, species : varchar(50), habitat : fregion)
air_pollution(id : int, pollutant : varchar(50), area : fregion)
```

4 Geometric Operations on Fuzzy Spatial Objects

This section defines FUSA operations according to the operation categories listed in Table 2. It includes operations returning crisp spatial objects (Section 4.1), fuzzy geometric set operations (Section 4.3), type-dependent geometric operations (Section 4.4), and membership-based geometric

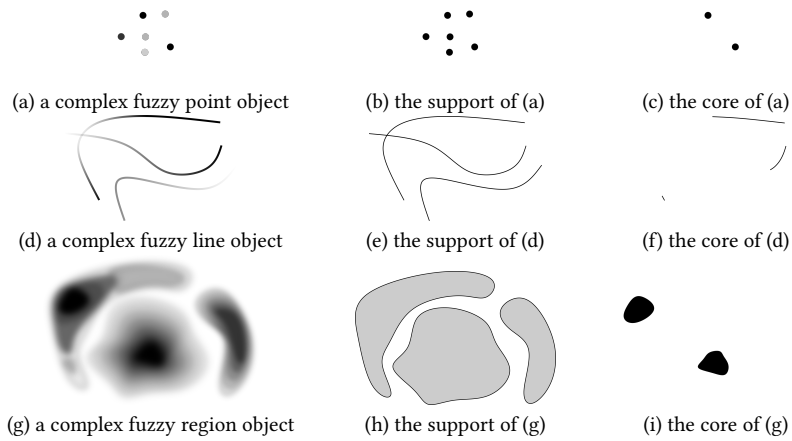


Fig. 14. Applying the operations *supp* and *core* on fuzzy spatial objects.

operations (Section 4.5). In addition, it includes some topology-based operations (Section 4.2). Examples of their application with regard to our running example are shown in Section 4.6.

In the following, let $\tilde{O}$ be a fuzzy spatial object of type $\alpha \in \{fpoint, fline, fregion\}$.

4.1 Operations Returning Crisp Spatial Objects

In this section, we specify operations that transform $\tilde{O}$ into a crisp spatial object. In general, the resulting crisp spatial object is formed by the points of $\tilde{O}$ that satisfy a filter condition based on their membership degrees.

Support. This operation, named *supp*, returns a crisp spatial object O that is the crisp counterpart of $\tilde{O}$. Hence, this operation has the overloaded signatures $fpoint \rightarrow point$, $fline \rightarrow line$, and $fregion \rightarrow region$. As shown in Sections 3.5–3.7, the application of *supp* to $\tilde{O}$ preserves the underlying geometric structure and properties of $\tilde{O}$ by extracting all the points of $\tilde{O}$ with membership degree greater than 0 to create the resulting crisp spatial object O. Figures 14(b), 14(e), and 14(h) show the results of applying the operation *supp* to the fuzzy spatial objects in Figures 14(a), 14(d), and 14(g), respectively. The operation *supp* is defined as

$supp(\tilde{O}) = O$ such that

(i) $O \in point$ if $\tilde{O} \in fpoint \vee O \in line$ if $\tilde{O} \in fline \vee O \in region$ if $\tilde{O} \in fregion$
(ii) $O = \{p \in \mathbb{R}^2 | \mu_{\tilde{O}}(p) > 0\}$

Core. As presented in Section 3.2.2, fuzzy set theory offers an operation called *core* that extracts all elements from a fuzzy set with membership degree equal to 1. However, directly applying this definition to a fuzzy spatial object (that is, $core(\tilde{O})$) could lead to a crisp point set that possibly does not correspond to an object of one of the crisp spatial data types due to geometric anomalies. For example, the core of a fuzzy line object can consist of its end points and some disjoint linear parts. Hence, we formally provide type-specific definitions of the core of a fuzzy spatial object. That is, we return the crisp point object, crisp line object, and crisp region object belonging to the core of a given a fuzzy spatial object. Figures 14(c), 14(f), and 14(i) depict examples of the operation *core*.

For this, we need two auxiliary operations. The first auxiliary operation is named *extract* and has the signature $\mathcal{P}(\mathbb{R}^2) \rightarrow point \times line \times region$. This function extracts three crisp spatial objects from a given point set as follows. Let $A \in \mathcal{P}(\mathbb{R}^2)$ and $extract(A) = (P, L, R)$. First, we determine the

largest region object contained in A. If $A \in region$, the operation *extract* terminates by returning $R = A$, $L = \perp_l$, and $P = \perp_p$ where $\perp_l$ and $\perp_p$ are the empty line object and the empty point object, respectively. Otherwise, we identify the set $A_{region} \subset A$ such that holds $A_{region} \in region$ and $\forall A_{region} \subset B \subseteq A : B \notin region$. We identify A_{region} by regularizing A, that is, $A_{region} = cl_{T_2^u}(int_{T_2^u}(A))$ (see Figure 5). This means that A_{region} can be either the empty region object $\perp_r$ or the largest region object that is properly contained in A. In any case, this means that $R = A_{region}$. Next, we obtain the set $A_{point} = \{q \in A \mid \exists \epsilon > 0 : N(q, \epsilon) - \{q\} = \emptyset\}$. That is, each point q of A_{point} is an isolated point of A since a neighborhood around q exists that does not contain any other point of A. If $A_{point} = \emptyset$, we set $P = \perp_p$. If A_{point} is infinite, then $A_{point} \notin point$ holds since an object of the spatial data type *point* may only contain a finite number of points. We can then only ignore A_{point} and set $P = \perp_p$. Otherwise, we set $P = A_{point}$.

We conclude that $B = A - int_{T_2^u}(A_{region}) - A_{point} \in line$ holds. The components of B are non-self-intersecting lines [71] that possibly share boundary points. Let $M = \{q \in B \mid degree(q) \geq 3\}$ where the function *degree* returns the number of line components that contain a given point q. Thus, M contains boundary points that are shared by at least three line components. Points with a degree equal to 2 denote the end of closed lines representing the boundaries of the components of A_{region}. We are not interested in them since we aim to form a crisp line object that possibly shares boundary points with the boundary of the crisp region object. This crisp line object is given as $A_{line} = B - bd_{T_2^u}(A_{region}) \cup M \in line$. This means that A_{line} can be either the empty crisp line object $\perp_l$ or a crisp line object that possibly shares single boundary points with the boundary of A_{region}. We set $L = A_{line}$.

The second auxiliary operation is named *project* and has the signature $point \times line \times region \times text \rightarrow \omega$ with $\omega \in \{point, line, region\}$. It retrieves one of the three components of a triple (P, L, R) according to the specified data type given as the last parameter. Hence, the following function calls and results are possible: (i) $project((P, L, R), \text{“}point\text{”}) = P$, (ii) $project((P, L, R), \text{“}line\text{”}) = L$, and (iii) $project((P, L, R), \text{“}region\text{”}) = R$.

We define the three type-specific operations $core_p$, $core_l$, and $core_r$ to yield a requested point, line, or region component of the core of a fuzzy spatial object $\tilde{O}$. They have the signatures (i) $core_p : \alpha \rightarrow point$, (ii) $core_l : \alpha \rightarrow line$, and (iii) $core_r : \alpha \rightarrow region$. The strategy behind the definition of these functions is as follows. First, we extract a crisp point set that corresponds to all the points with membership degree equal to 1 from $\tilde{O}$. Then, we invoke our auxiliary operation *extract* for uniquely transforming the crisp point set into valid crisp spatial objects. Finally, we capture the desired crisp spatial object by using the auxiliary operation *project*. But the result of the projection can be an empty crisp spatial object, according to the underlying definition of *extract*. Further, we are able to determine cases in which it is impossible to return the linear component or the areal component of the core of $\tilde{O}$ because of the lower dimension of its data type. We are now able to give the definitions for $core_p$, $core_l$, and $core_r$ as:

$$core_p(\tilde{O}) = project(extract(\{p \in \mathbb{R}^2 \mid \mu_{\tilde{O}}(p) = 1\}), \text{“}point\text{”})$$

$$core_l(\tilde{O}) = \begin{cases} \perp_l & \text{if } \tilde{O} \in fpoint \\ project(extract(\{p \in \mathbb{R}^2 \mid \mu_{\tilde{O}}(p) = 1\}), \text{“}line\text{”}) & \text{otherwise} \end{cases}$$

$$core_r(\tilde{O}) = \begin{cases} \perp_r & \text{if } \tilde{O} \in fpoint \vee \tilde{O} \in fline \\ project(extract(\{p \in \mathbb{R}^2 \mid \mu_{\tilde{O}}(p) = 1\}), \text{“}region\text{”}) & \text{if } \tilde{O} \in fregion \end{cases}$$

Alpha-cut and Strict Alpha-cut. Similarly to the operation *core*, the direct application of the operations *alpha-cut* and *strict-alpha-cut* from fuzzy set theory (Section 3.2.2) to $\tilde{O}$ could lead to a crisp point set that does possibly not represent a valid crisp spatial object. We specify the

three type-specific signatures for the operation *alpha-cut* as (i) $alpha\text{-}cut_p : \alpha \times [0,1] \rightarrow point$, (ii) $alpha\text{-}cut_l : \alpha \times [0,1] \rightarrow line$, and (iii) $alpha\text{-}cut_r : \alpha \times [0,1] \rightarrow region$. The three corresponding signatures for the operation *strict-alpha-cut* are similar but require a replacement of the closed interval $[0,1]$ by the right-open interval $[0,1[$. Let $a \in [0,1]$. By using a similar strategy as for the definition of the operation *core*, we define the three operations for the operation *alpha-cut* as follows:

$$alpha\text{-}cut_p(\tilde{O}, a) = project(extract(\{p \in \mathbb{R}^2 \mid \mu_{\tilde{O}}(p) \geq a\}), \text{“}point\text{”})$$

$$alpha\text{-}cut_l(\tilde{O}, a) = \begin{cases} \perp_l & \text{if } \tilde{O} \in fpoint \\ project(extract(\{p \in \mathbb{R}^2 \mid \mu_{\tilde{O}}(p) \geq a\}), \text{“}line\text{”}) & \text{otherwise} \end{cases}$$

$$alpha\text{-}cut_r(\tilde{O}, a) = \begin{cases} \perp_r & \text{if } \tilde{O} \in fpoint \ \vee \tilde{O} \in fline \\ project(extract(\{p \in \mathbb{R}^2 \mid \mu_{\tilde{O}}(p) \geq a\}), \text{“}region\text{”}) & \text{if } \tilde{O} \in fregion \end{cases}$$

The operation *strict-alpha-cut* is defined similarly by replacing the symbol '$\geq$' by '$>$' and by selecting $a \in [0,1[$.

Note that $\forall \gamma \in \{p, l, r\} : alpha\text{-}cut_\gamma(\tilde{O}, 1) = core_\gamma(\tilde{O})$ holds. Further, we can derive (i) $\forall \tilde{O} \in fpoint : strict\text{-}alpha\text{-}cut_p(\tilde{O}, 0) = supp(\tilde{O})$, (ii) $\forall \tilde{O} \in fline : strict\text{-}alpha\text{-}cut_l(\tilde{O}, 0) = supp(\tilde{O})$, and (iii) $\forall \tilde{O} \in fregion : strict\text{-}alpha\text{-}cut_r(\tilde{O}, 0) = supp(\tilde{O})$.

Range. The operation *range* allows us to yield a crisp spatial object composed of the points of $\tilde{O}$ that have membership degrees of a given open or closed interval. Let $a, b \in [0,1]$ with $a \leq b$. They specify the range of the interval. Let $\phi_l, \phi_r \in \theta = \{<, \leq\}$. They determine whether the left and right sides of the interval are open ($<$) or closed ($\leq$). Hence, the quadruple (a, b, ϕ_l, ϕ_r) allows us to define closed intervals, left open intervals, right open intervals, and open intervals. For example, $[0.5, 0.7]$ is represented as $(0.5, 0.7, \leq, \leq)$, $]0.5, 0.7]$ as $(0.5, 0.7, <, \leq)$, $[0.5, 0.7[$ as $(0.5, 0.7, \leq, <)$, and $]0.5, 0.7[$ as $(0.5, 0.7, <, <)$. We specify the following three type-specific signatures: (i) $range_p : \alpha \times [0,1] \times [0,1] \times \theta \times \theta \rightarrow point$, (ii) $range_l : \alpha \times [0,1] \times [0,1] \times \theta \times \theta \rightarrow line$, and (iii) $range_r : \alpha \times [0,1] \times [0,1] \times \theta \times \theta \rightarrow region$. We now are able to define these operations as follows:

$$range_p(\tilde{O}, a, b, \phi_l, \phi_r) = project(extract(\{p \in \mathbb{R}^2 \mid a \ \phi_l \ \mu_{\tilde{O}}(p) \ \phi_r \ b\}), \text{“}point\text{”})$$

$$range_l(\tilde{O}, a, b, \phi_l, \phi_r) = \begin{cases} \perp_l & \text{if } \tilde{O} \in fpoint \\ project(extract(\{p \in \mathbb{R}^2 \mid a \ \phi_l \ \mu_{\tilde{O}}(p) \ \phi_r \ b\}), \text{“}line\text{”}) & \text{otherwise} \end{cases}$$

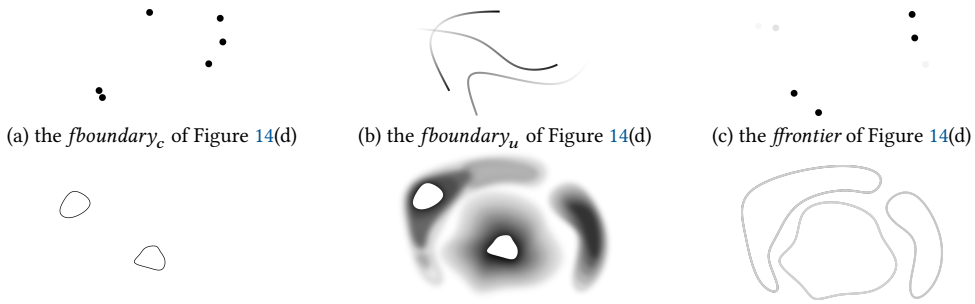
$$range_r(\tilde{O}, a, b, \phi_l, \phi_r) = \begin{cases} \perp_r & \text{if } \tilde{O} \in fpoint \ \vee \\ & \tilde{O} \in fline \\ project(extract(\{p \in \mathbb{R}^2 \mid a \ \phi_l \ \mu_{\tilde{O}}(p) \ \phi_r \ b\}), \text{“}region\text{”}) & \text{if } \tilde{O} \in fregion \end{cases}$$

The operation *range* generalizes the previously defined operations *supp*, *core*, *alpha-cut*, and *strict alpha-cut*. The latter operations can be redefined as follows:

$$\forall \tilde{O} \in fpoint : supp(\tilde{O}) = range_p(\tilde{O}, 0, 1, <, \leq)$$
$$\forall \tilde{O} \in fline : supp(\tilde{O}) = range_l(\tilde{O}, 0, 1, <, \leq)$$
$$\forall \tilde{O} \in fregion : supp(\tilde{O}) = range_r(\tilde{O}, 0, 1, <, \leq)$$
$$\forall \gamma \in \{p, l, r\} : core_\gamma(\tilde{O}) = range_\gamma(\tilde{O}, 1, 1, \leq, \leq)$$
$$\forall \gamma \in \{p, l, r\} : alpha\text{-}cut_\gamma(\tilde{O}, a) = range_\gamma(\tilde{O}, a, 1, \leq, \leq)$$
$$\forall \gamma \in \{p, l, r\} : strict\text{-}alpha\text{-}cut_\gamma(\tilde{O}, a) = range_\gamma(\tilde{O}, a, 1, <, \leq)$$

4.2 Topology-based Operations

As discussed in Section 3.2.3, the notions of fuzzy boundary and fuzzy frontier of a fuzzy set allow us to provide two different concepts of the borders of fuzzy spatial objects. However, directly



(a) the *fboundary*$_c$ of Figure 14(d) (b) the *fboundary*$_u$ of Figure 14(d) (c) the *ffrontier* of Figure 14(d)

(d) the *fboundary*$_c$ of Figure 14(g) (e) the *fboundary*$_u$ of Figure 14(g) (f) the *ffrontier* of Figure 14(g)

Fig. 15. Applying the operations *fboundary*$_c$, *fboundary*$_u$, and *ffrontier* on the fuzzy line and region objects of Figures 14(d) and 14(g).

applying these notions to a fuzzy spatial object could lead to a fuzzy point set that possibly violates the definitions of our fuzzy spatial data types. Hence, we define these operations in a different way.

Fuzzy Boundary. Intuitively, we combine the notion of the fuzzy boundary of a fuzzy set (Section 3.2.3) with the notion of the boundary of a crisp spatial object (e.g., defined in [71]). The fuzzy boundary of $\tilde{O}$ consists of two components, the first one representing the crisp boundary of the core, and the second one representing the fuzzy non-core part of $\tilde{O}$. We define two operations named *fboundary*$_c$ and *fboundary*$_u$ to extract the two components of the fuzzy boundary of $\tilde{O}$. We further define that if $\tilde{O}$ is an *fpoint* object, then both of its boundary components are empty. This is compatible with the definition of the crisp boundary of crisp point objects. If $\tilde{O}$ is an *fline* object, then its fuzzy boundary parts are formed by an *fpoint* object and an *fline* object. Further, if $\tilde{O}$ is an *fregion* object, then its fuzzy boundary parts are made up of an *fline* object and an *fregion* object. Figures 15(a), 15(b), 15(d), and 15(e) depict the two kinds of fuzzy boundaries of the fuzzy spatial objects of Figures 14(d) and 14(g). To define the operation *fboundary*$_c$, we make use of the operation *core*. For the operation *fboundary*$_u$, we employ the operation *range* (Section 4.2), which yields a crisp spatial object. The next step is then to associate each point of this object to the corresponding membership degree of $\tilde{O}$. Let $\tilde{\perp}_p$ be the empty fuzzy point object. We define *fboundary*$_c$ and *fboundary*$_u$ as follows:

$$fboundary_c(\tilde{O}) = \begin{cases} \tilde{\perp}_p & \text{if } \tilde{O} \in fpoint \\ \{(p,1) \in \tilde{O} \mid p \in bd_{T_2^u}(core_l(\tilde{O}))\} \in fpoint & \text{if } \tilde{O} \in fline \\ \{(p,1) \in \tilde{O} \mid p \in bd_{T_2^u}(core_r(\tilde{O}))\} \in fline & \text{if } \tilde{O} \in fregion \end{cases}$$

$$fboundary_u(\tilde{O}) = \begin{cases} \tilde{\perp}_p & \text{if } \tilde{O} \in fpoint \\ \{(p,\mu_{\tilde{O}}(p)) \in \tilde{O} \mid p \in range_l(\tilde{O},0,1,<,<)\} \in fline & \text{if } \tilde{O} \in fline \\ \{(p,\mu_{\tilde{O}}(p)) \in \tilde{O} \mid p \in range_r(\tilde{O},0,1,<,<)\} \in fregion & \text{if } \tilde{O} \in fregion \end{cases}$$

Fuzzy Frontier. Intuitively, this operation, named *ffrontier*, applies another notion of the fuzzy boundary of a fuzzy set (Section 3.2.3) to a fuzzy spatial object $\tilde{O}$. The operation *ffrontier* on $\tilde{O}$ returns a fuzzy spatial object whose support corresponds to the boundary of the support of $\tilde{O}$. Figures 15(c) and 15(f) depict the fuzzy frontier of the fuzzy spatial objects of Figures 14(d) and 14(g). We define *ffrontier* as follows:

$$ffrontier(\tilde{O}) = \begin{cases} \tilde{\perp}_p & \text{if } \tilde{O} \in fpoint \\ \{(p,\mu_{\tilde{O}}(p)) \in \tilde{O} \mid p \in bd_{T_2^u}(supp(\tilde{O}))\} \in fpoint & \text{if } \tilde{O} \in fline \\ \{(p,\mu_{\tilde{O}}(p)) \in \tilde{O} \mid p \in bd_{T_2^u}(supp(\tilde{O}))\} \in fline & \text{if } \tilde{O} \in fregion \end{cases}$$

The definitions of fuzzy boundary and fuzzy frontier ensure the correspondence to the conventional boundary of crisp spatial objects. Let $\tilde{C}$ be a fuzzy spatial object containing points with membership degrees equal to 1 only, which corresponds to a crisp spatial object C. Then we have the following equalities:

$$supp(fboundary_c(\tilde{C})) = supp(ffrontier(\tilde{C})) = bd_{T_2^u}(C)$$

4.3 Fuzzy Geometric Set Operations

Fuzzy geometric set operations operate on two fuzzy spatial objects of the same fuzzy spatial data type and calculate a new fuzzy spatial object of the same fuzzy spatial data type as the result of their *fuzzy geometric union*, *fuzzy geometric intersection*, or *fuzzy geometric difference*. This means that their signature is $\psi : \alpha \times \alpha \rightarrow \alpha$ with $\psi \in \{funion, fintersection, fdifference\}$ and $\alpha \in \{fpoint, fline, fregion\}$. Section 4.3.1 provides a formal definition of our fuzzy geometric set operations. The semantics and expressiveness of fuzzy geometric set operations can be influenced and thus changed by a variety of different operators with special properties. The selection of suitable operators depends on the applications in which they are used and is therefore *context-dependent*. Technically this leads to *parameterized* fuzzy geometric set operations. Section 4.3.2 shows that all context-dependent fuzzy geometric set operations fulfill closure properties. This means that all fuzzy spatial objects that are the result of any context-dependent fuzzy geometric set operation fulfill all conditions of their respective fuzzy spatial data type definition.

4.3.1 Formal Definition of Context-Dependent Fuzzy Geometric Set Operations. When we compute the intersection, union, and difference of two fuzzy spatial objects $\tilde{A}, \tilde{B} \in \alpha$, the membership degrees of $\tilde{A}$ and $\tilde{B}$ are combined at each location $p \in \mathbb{R}^2$ by a context-dependent operator that implements a particular interpretation of spatial fuzziness. The standard operators that we will assume for the three operations are the same as those defined for the fuzzy set operations on fuzzy sets (Section 3.2.2), that is, the operators *max* for fuzzy (geometric) union, *min* for fuzzy (geometric) intersection, and '−' for fuzzy (geometric) difference. But fuzzy set theory offers a large variety of alternative interpretations provided by different operators for this purpose that are also interesting in the spatial domain. In case of fuzzy (geometric) intersection, they are summarized by the concept of *t-norms* (*triangular norms*) that possess particular properties, are independent of the values $p \in \mathbb{R}^2$, and only depend on the values $\mu_{\tilde{A}}(p), \mu_{\tilde{B}}(p) \in [0, 1]$ [41]. Therefore, we may ignore the parameter p and assume that the arguments of a t-norm are arbitrary numbers $a, b \in [0, 1]$. This leads to the definition that a *t-norm* T is a binary function $T : [0, 1] \times [0, 1] \rightarrow [0, 1]$ with the following properties for all $a, b, c \in [0, 1]$:

(i)	$T(a, 1) = a$	(*boundary condition*, *identity element*)
(ii)	$T(a, b) = T(b, a)$	(*commutativity*)
(iii)	$b \leq c$ implies $T(a, b) \leq T(a, c)$	(*monotonicity*)
(iv)	$T(a, T(b, c)) = T(T(a, b), c)$	(*associativity*)

Condition (i) ensures that the fuzzy set intersection becomes the classical set intersection when the interval $[0, 1]$ is reduced to the crisp set $\{0, 1\}$ since we obtain $T(0, 1) = 0$ and $T(1, 1) = 1$. Condition (ii) states that a t-norm, and hence fuzzy intersection, is symmetric, that is, indifferent to the order in which the sets to be combined are considered. Further, we can conclude that $T(1, 0) = 0$ holds. Condition (iii) expresses that a decrease in the degree of membership cannot lead to an increase in the degree of membership in the intersection. Incorporating condition (ii), T is non-decreasing in both arguments. Further, we can see that $0 \leq 1$ implies $T(0, 0) \leq T(0, 1) = 0$ and $T(0, 0) = 0$. Condition (iv) allows us to apply a t-norm to and thus determine the intersection of any number of sets in any order of pairwise grouping desired; this enables us to extend the fuzzy intersection operation to more than two sets.

Table 4. Examples of t-norms and t-conorms that can be Deployed as Membership Functions for the Fuzzy Geometric Intersection and the Fuzzy Geometric Union Operation, Respectively (Let $a, b \in [0, 1]$)

t-norm		t-conorm	
Standard intersection, minimum t-norm, Gödel t-norm	$T_{mi}(a, b) = \min(a, b)$	*Standard union, maximum t-conorm*	$C_{ma}(a, b) = \max(a, b)$
Bounded difference, Lukasiewicz t-norm	$T_{bd}(a, b) = \max(0, a + b - 1)$	*Bounded sum*	$C_{bs}(a, b) = \min(a + b, 1)$
Algebra product, product t-norm	$T_{ap}(a, b) = ab$	*Algebraic sum, probabilistic sum*	$C_{as}(a, b) = a + b - ab$
Drastic intersection, drastic t-norm	$T_{di}(a, b) = \begin{cases} \min(a, b) & \text{if } \max(a, b) = 1 \\ 0 & \text{otherwise} \end{cases}$	*Drastic union, drastic t-conorm*	$C_{du}(a, b) = \begin{cases} \max(a, b) & \text{if } \min(a, b) = 0 \\ 1 & \text{otherwise} \end{cases}$
Nilpotent minimum	$T_{nm}(a, b) = \begin{cases} \min(a, b) & \text{if } a + b > 1 \\ 0 & \text{otherwise} \end{cases}$	*Nilpotent maximum*	$C_{nm}(a, b) = \begin{cases} \max(a, b) & \text{if } a + b < 1 \\ 1 & \text{otherwise} \end{cases}$
Yager t-norms	$T_{ya}(a, b) = 1 - \min(1, [(1 - a)^w + (1 - b)^w]^{1/w}), w > 0$	*Yager t-conorms*	$C_{ya}(a, b) = \min(1, (a^w + b^w)^{1/w}), w > 0$
Schweizer & Sklar t-norms	$T_{ss}(a, b) = (\max(0, a^p + b^p - 1))^{1/p}, p \neq 0$	*Schweizer & Sklar t-conorms*	$C_{ss}(a, b) = 1 - (\max(0, (1 - a)^p + (1 - b)^p - 1))^{1/p}, p \neq 0$
Hamacher product	$T_{hp}(a, b) = \begin{cases} 0 & \text{if } a = b = 0 \\ \frac{ab}{a+b-ab} & \text{otherwise} \end{cases}$	*Einstein sum*	$C_{es}(a, b) = \frac{a+b}{1+ab}$

The left column of Table 4 shows some t-norms that can be used as operators in membership functions for fuzzy (geometric) intersection (each defined for all $a, b \in [0, 1]$). Geoscientists could make use of one of these t-norms, or employ other t-norms proposed in the literature, or invent their own t-norms tailored to the spatial application. We define a set *tnorm* that contains the function names of all listed t-norms:

$$tnorm = \{T_{mi}, T_{bd}, T_{ap}, T_{di}, T_{nm}, T_{ya}, T_{ss}, T_{hp}\}$$

In case of fuzzy (geometric) union, operators are deployed that are called *t-conorms* (*triangular conorms*) and that are dual to t-norms under the order-reversing operation that assigns $1 - x$ to x on the interval $[0, 1]$. Given a t-norm T and its complementary t-conorm C, we obtain the following two dual relationships for all $a, b \in [0, 1]$:

$$T(a, b) = 1 - C(1 - a, 1 - b)$$
$$C(a, b) = 1 - T(1 - a, 1 - b)$$

It follows that a t-conorm satisfies similar conditions like a t-norm. These conditions can be used for an equivalent definition of t-conorms independently of t-norms. A *t-conorm* C is a function $C : [0, 1] \times [0, 1] \rightarrow [0, 1]$ with the following four properties for all $a, b, c \in [0, 1]$:

(i) $C(a, 0) = a$ (*boundary condition, identity element*)
(ii) $C(a, b) = C(b, a)$ (*commutativity*)
(iii) $b \leq c$ implies $C(a, b) \leq C(a, c)$ (*monotonicity*)
(iv) $C(a, C(b, c)) = C(C(a, b), c)$ (*associativity*)

A comparison of these conditions with the conditions for t-norms shows that they only differ in the boundary condition. Conditions (i) to (iii) ensure that the fuzzy set union becomes the classical set union when the interval $[0, 1]$ is reduced to the crisp set $\{0, 1\}$. We obtain $C(0, 0) = 0$,

$C(0,1) = C(1,0) = C(1,1) = 1$. Otherwise, the conditions have the same rationale as those for t-norms.

The right column of Table 4 shows the dual t-conorms of the respective t-norms in the left column. They can be used as operators in membership functions for fuzzy (geometric) union. We define a set *tconorm* that contains the function names of all listed t-conorms:

$$tconorm = \{C_{ma}, C_{bs}, C_{as}, C_{du}, C_{nm}, C_{ya}, C_{ss}, C_{es}\}$$

In case of fuzzy (geometric) difference, possible operators are the standard difference $-$, the fuzzy bounded difference $\dot{-}$, the fuzzy symmetric difference Δ, and the fuzzy absolute difference $|-|$. Their definitions have been given in Section 3.2.2. We collect these alternatives in the set

$$diff = \{-, \dot{-}, \Delta, |-|\}$$

A suitable selection of an operator from the sets *tnorm*, *tconorm*, or *diff* rests on several criteria. First, a *justification* of the selection is needed and ranges from intuitive argumentation to empirical or axiomatic justification. A selection depends on the geospatial application in which it is deployed and must be appropriate for the domain where it is used. Therefore, such a selection is *context-dependent*. Second, if some operators share several characteristics, *axiomatic strength* should decide the selection process in the sense that the operator with the less restrictive set of properties is chosen. Third, *generality* and *adaptability* play a role and range from uniquely defined, unadaptable concepts via parameterized families of membership functions (for example, Yager, Schweizer and Sklar) to general classes of membership functions. Fourth, *computational performance* regarding numerical operations is important. It is obvious that the *max* and *min* functions require less computational effort than parameterized functions. To ensure a good numerical efficiency, the first kind of functions should be preferred. Fifth, a sometimes desirable property is *compensation*. If $T(a,b) = z$ with $a, b, z \in [0,1]$, T is *compensatory* if, whenever modifying a to $a' \in [0,1]$, it can be achieved that $T(a', b') = z$ for a value $b' \in [0,1]$ as a modification of b. For example, if $a = 0.3$ and $b = 0.4$ and the operator T is *min*, we obtain $T(a,b) = \min(0.3, 0.4) = 0.3 = z$. If we select $a' = 0.2$, no matter what kind of value we would select for b', we could not achieve that $\min(a', b') = 0.3$ since $\min(0.2, b') \leq 0.2$; hence, the operator *min* is not compensatory. However, if T is the algebraic product, we obtain $0.3 \cdot 0.4 = 0.12 = z$. If we select $a' = 0.2$, we have $0.2 \cdot b' = 0.12$. This leads to $b' = 0.12/0.2 = 0.6$; hence, the algebraic product operator is compensatory.

Besides t-norms, t-conorms, and *diff* operators, we provide the two additional special operators *first*, *second* $: [0,1] \times [0,1] \rightarrow [0,1]$ that give exclusive priority either to their first or second argument. They are defined for all $a, b \in [0,1]$ as $first(a,b) = a$ and $second(a,b) = b$. For example, assume that a polluted river modeled as a fuzzy line flows across a city modeled as a fuzzy region, and an environmental science expert wants to know the length of the river inside this city in order to calculate the amount of decontamination liquid that needs to be poured into the river to cope with the polluted water. Then the membership values of the fuzzy line are relevant and should be preserved while the membership values of the fuzzy region can be ignored. Hence, the function *first* would be used in a fuzzy geometric intersection operation.

For the operations *fintersection*, *funion*, and *fdifference*, we introduce the three operator types *itype* (intersection type), *utype* (union type), and *dtype* (difference type). Depending on a selected operator name, the corresponding operator is called in the membership function of one of the three fuzzy geometric set operations.

(i) $itype = tnorm \cup \{first, second\}$
(ii) $utype = tconorm \cup \{first, second\}$
(iii) $dtype = diff \cup \{first, second\}$

This leads to generalized fuzzy geometric set operations that are parameterized with a t-norm, t-conorm, *diff* function, *first* function, or *second* function respectively and have the following signatures ($\alpha \in \{fpoint, fline, fregion\}$):

(iii) $fintersection: \ \alpha \times \alpha \times itype \ \rightarrow \ \alpha$
(iii) $funion: \ \alpha \times \alpha \times utype \ \rightarrow \ \alpha$
(iii) $fdifference: \ \alpha \times \alpha \times dtype \ \rightarrow \ \alpha$

For given $\tilde{A}, \tilde{B} \in \alpha$, $f \in itype$, $g \in utype$, and $h \in dtype$, this means that

(i) $fintersection(\tilde{A}, \tilde{B}, f) \ = \ reg_\alpha(\tilde{A} \cap_f \tilde{B})$
(ii) $funion(\tilde{A}, \tilde{B}, g) \ = \ reg_\alpha(\tilde{A} \cup_g \tilde{B})$
(iii) $fdifference(\tilde{A}, \tilde{B}, h) \ = \ reg_\alpha(\tilde{A} -_h \tilde{B})$

where

(i) $\tilde{A} \cap_f \tilde{B} \ = \ \{(x, \mu_{\tilde{A} \cap_f \tilde{B}}(x)) \mid x \in \mathbb{R}^2 \wedge \mu_{\tilde{A} \cap_f \tilde{B}}(x) = f(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))\}$
(ii) $\tilde{A} \cup_g \tilde{B} \ = \ \{(x, \mu_{\tilde{A} \cup_g \tilde{B}}(x)) \mid x \in \mathbb{R}^2 \wedge \mu_{\tilde{A} \cup_g \tilde{B}}(x) = g(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))\}$
(iii) $\tilde{A} -_h \tilde{B} \ = \ \{(x, \mu_{\tilde{A} -_h \tilde{B}}(x)) \mid x \in \mathbb{R}^2 \wedge \mu_{\tilde{A} -_h \tilde{B}}(x) = h(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))\}$

The function reg_α ensures closure properties of the fuzzy geometric operations and will be explained in Section 4.3.2.

4.3.2 Closure Properties of Fuzzy Geometric Set Operations. In Sections 3.5 to 3.7 we have seen that especially fuzzy line objects and fuzzy region objects are not arbitrary fuzzy point sets but are defined in a way so that they reflect their smooth and anomaly-free nature in spatial reality and in geospatial applications. In this sense, fuzzy line objects represent geometrically smooth curves and express their spatial fuzziness by (finitely upper semi-)continuous membership functions. Fuzzy region objects do not allow any kind of anomalies in their geometries and express their spatial fuzziness by the same kind of membership functions. In both cases, a finite number of continuity gaps is allowed in the membership functions. But applying the intersection, union, and difference operations on fuzzy sets to two fuzzy line objects or two fuzzy region objects respectively usually introduces geometric anomalies so that one does not get a fuzzy line object or a fuzzy region object respectively as a result. An example is the classical fuzzy intersection of two fuzzy line objects that can result in a number of single fuzzy points, which are considered an anomaly here, and in a number of simple fuzzy lines. In other words, our fuzzy spatial data types *fline* and *fregion* are generally not closed under the classical fuzzy set operations. Hence, the resulting fuzzy point sets have to be "repaired" or *regularized*. To this end, we apply a type-specific *regularization function* reg_α to the results of the fuzzy set operations in the definitions of our fuzzy geometric set operations, as introduced in Section 4.3.1.

Let $\tilde{A}, \tilde{B} \in \alpha$. For all three combinations of fuzzy spatial objects $\tilde{A}$ and $\tilde{B}$ of the same fuzzy spatial data type α, let $\tilde{C}$ be the fuzzy point set with $\tilde{C} = \tilde{A} \cap_f \tilde{B}$ or $\tilde{C} = \tilde{A} \cup_g \tilde{B}$ or $\tilde{C} = \tilde{A} -_h \tilde{B}$. Note that $\tilde{C}$ is not necessarily a valid fuzzy spatial object. In the following, we describe the required regularization process on $\tilde{C}$.

The fuzzy spatial data type *fpoint* is closed under the fuzzy set operations since its objects consist of isolated single fuzzy points to which the fuzzy set operations are applied that yield isolated single fuzzy points again as a result. Therefore, $reg_{fpoint} = id$ holds where $id : Z \rightarrow Z$ is the *identity function* on some set Z with $id(x) = x$ for all $x \in Z$. Using the generalization that $id(Z) = \{id(x) \mid x \in Z\}$, we get $reg_{fpoint}(\tilde{C}) = id(\tilde{C}) = \tilde{C}$.

In the case of two fuzzy line objects $\tilde{A}$ and $\tilde{B}$, in a first step, we collect all linear parts of $\tilde{C}$ by ignoring their membership degrees in a crisp line object Y as follows. We define $\mathcal{M}(X) = \{\tilde{A} \mid \tilde{A} = \{(x, \mu_{\tilde{A}}(x)) \mid x \in X\}, \mu_{\tilde{A}} : X \rightarrow [0, 1]\}$ as the set of all fuzzy sets for a given universe X.

Y is then determined by the function *get_cline* with the signature $get_cline : \mathcal{M}(\mathbb{R}^2) \rightarrow line$ as $Y = get_cline(\tilde{C}) = project(extract(supp(\tilde{C})), \text{“}line\text{”})$. This means we obtain a geometric structure without anomalies that conforms to a valid object of type *line*. Y can be decomposed into a finite set L of simple crisp line objects, that is, $L = \{l_1, \ldots, l_n\}$ for some $n \in \mathbb{N}$ such that $Y = \bigcup_{i=1}^{n} l_i$. We assume a function *get_comps* with the signature $get_comps : line \rightarrow \mathcal{P}(line)$ for this purpose. By definition, each $l \in L$ is given by a function f_l that models its continuous simple crisp curve. Since each point p in l has a membership degree in $\tilde{C}$, we can create a fuzzy point set $\tilde{l}$ by annotating p with its corresponding membership degree in $\tilde{C}$. This is achieved by applying the function $annotate : line \rightarrow \mathcal{M}(\mathbb{R}^2)$ to l and obtaining $annotate(l) = \tilde{l} = \{(p, \mu_{\tilde{l}}(p)) \mid p \in f_l([0,1]) \wedge \mu_{\tilde{l}}(p)) = \mu_{\tilde{C}}(p))\}$. We extend the function *annotate* to L and obtain $L_f = annotate(L) = \{annotate(l) \mid l \in L\} = \{\tilde{l}_1, \ldots, \tilde{l}_n\}$.

The next step is to eliminate isolated discontinuities in the transitions of the membership degrees that annotate the points in each fuzzy point set $\tilde{l} \in L_f$. For each $\tilde{l} \in L_f$, we build a fuzzy set $\tilde{E}$ with the membership function $\mu_{\tilde{E}}$ from the composition of $\mu_{\tilde{l}}$ and f_l so that this composition maps a value $s \in [0,1]$ to $\mu_{\tilde{l}}(f_l(s)) \in [0,1]$, similarly to the conditions (iii) and (iv) in the formal definition of simple fuzzy lines (Section 3.6). Then, we apply the closure and interior operations on $\tilde{E}$, that is, $\tilde{E} = cl_{\mathcal{T}_1^u}(int_{\mathcal{T}_1^u}(\tilde{E}))$. Thus, $\mu_{\tilde{E}}$ assigns membership degrees to the points of $\tilde{l}$ without isolated discontinuities. To perform this step, we define the function *reg_sflines*, with signature $reg_sflines : \mathcal{M}(\mathbb{R}^2) \rightarrow fline$, for $\tilde{l} \in L_f$ as $\tilde{l} = reg_sfline(\tilde{l}) = \{(p, \mu_{\tilde{E}}(p)) \mid p \in f_l([0,1]) \wedge \mu_{\tilde{E}} = \mu_{\tilde{l}} \circ f_l \wedge \tilde{E} = cl_{\mathcal{T}_1^u}(int_{\mathcal{T}_1^u}(\tilde{E}))\}$. This regularization process turns $\tilde{l}$ into a valid simple fuzzy line object by eliminating isolated discontinuities. We then extend this function to L_f to obtain a new version of L_f, that is, $L_f = reg_sflines(L_f) = \{reg_sflines(\tilde{l}) \mid \tilde{l} \in L_f\}$.

The final step is to compute the fuzzy union of all simple fuzzy line objects in L_f to obtain a valid complex fuzzy line object $\tilde{Y}$. We determine $\tilde{Y}$ by using the function *agg_funion* with signature $agg_funion : \mathcal{P}(fline) \rightarrow fline$ and define $\tilde{Y} = agg_funion(L_f) = \bigcup_{i=1}^{n} \tilde{l}_i$ with $\tilde{l}_i \in L_f$. We can apply the fuzzy union operation because all simple fuzzy lines are regularized and their shared end points have the same membership degree. This means that the role of the fuzzy union is to aggregate all simple fuzzy lines in a unique geometric structure since in this case it does not change the membership degrees of shared points. The resulting $\tilde{Y}$ fulfills all requirements of the *fline* definition. First, the supports of any two simple fuzzy lines belonging to $\tilde{Y}$ are either disjoint or adjacent in single end points since the underlying geometric structure of $\tilde{Y}$ is Y, which is a valid complex crisp line object. Second, $\tilde{Y}$ is consistent in the sense that the common end points of its simple fuzzy lines have the same membership degree. This is due to the fact that the operator used in the fuzzy geometric set operation guarantees that shared points have the same membership degree. This also holds after repairing isolated discontinuities in the shared points. By considering the aforementioned functions, we get $reg_{fline}(\tilde{C}) = agg_funion(reg_sflines(annotate(get_comps(get_cline(\tilde{C})), \tilde{C})))$.

Also in the case of two fuzzy region objects $\tilde{A}$ and $\tilde{B}$, $\tilde{C}$ can be afflicted with anomalies regarding the resulting object geometry and its membership degrees. Geometric anomalies have been described at the beginning of this section as well as in Section 3.7. We eliminate them in a similar way as the steps employed to regularize a fuzzy point set to a valid fuzzy line object. First, we collect all areal parts of $\tilde{C}$ in a crisp region object W by using the function *get_cregion* with signature $get_cregion : \mathcal{M}(\mathbb{R}^2) \rightarrow region$. We determine W as $W = get_cregion(\tilde{C}) = project(extract(supp(\tilde{C}), \text{“}region\text{”}))$. We can overload the function *get_comps* to the signature $get_comps : region \rightarrow \mathcal{P}(region)$ to decompose W into a finite set R of simple region objects, that is, $R = \{r_1, \ldots, r_n\}$ for some $n \in \mathbb{N}$ such that $R = \bigcup_{i=1}^{n} r_i$. By definition, each $r \in R$ is a bounded and connected set and is free from geometric anomalies. We create a fuzzy point set $\tilde{r}$ by labeling each point p in r with its corresponding membership degree in $\tilde{C}$. For this, we extend the function *annotate* to $annotate$:

region $\rightarrow \mathcal{M}(\mathbb{R}^2)$. It results in $annotate(r) = \tilde{r} = \{(p, \mu_{\tilde{r}}(p)) \mid p \in r \wedge \mu_{\tilde{r}}(p)) = \mu_{\tilde{C}}(p))\}$ if applied to r. Further, it yields $R_f = annotate(R) = \{annotate(r) \mid r \in R\} = \{\tilde{r}_1, \ldots, \tilde{r}_n\}$ when extending the function *annotate* to R. We then have to eliminate isolated discontinuities in the transitions of the membership degrees in each fuzzy point set $\tilde{r} \in R_f$. To this end, we apply the closure and interior operations on $\tilde{r}$ by employing the function $reg_sfregions : \mathcal{M}(\mathbb{R}^2) \rightarrow fregion$ defined as $\tilde{r} = reg_sfregions(\tilde{r}) = cl_{\mathcal{T}_2^u}(int_{\mathcal{T}_2^u}(\tilde{r}))$. By extending this function to R_f, we obtain a new version of R_f, that is, $R_f = reg_sfregions(R_f) = \{reg_sfregions(\tilde{r}) \mid \tilde{r} \in R_f\}$. Finally, we compute the fuzzy union of all simple fuzzy region objects in R_f to obtain a valid complex fuzzy region object $\tilde{W}$. For this, we extend the function *agg_funion* to $agg_funion : \mathcal{P}(fregion) \rightarrow fregion$, which is defined as $\tilde{W} = agg_funion(R_f) = \bigcup_{i=1}^{n} \tilde{r}_i$ with $\tilde{r}_i \in R_f$. Similarly to the regularization of fuzzy lines, we can apply the classical fuzzy union operation because all simple fuzzy regions are regularized and their shared boundary points have the same membership degree. As a result, $\tilde{W}$ conforms to the *fregion* definition since its geometric structure does not have anomalies and the distribution of the membership degrees does not have isolated discontinuities. Therefore, we get $reg_{fregion}(\tilde{C}) = agg_funion(reg_sfregions(annotate(get_cregion(\tilde{C})), \tilde{C}))$.

4.4 Type-Dependent Geometric Operations

In this section, we define fuzzy geometric operations that can be applied to specific fuzzy spatial data types only. This means that they are type-dependent and yield another fuzzy spatial object as output.

Fuzzy Common Points. This operation, named *fcommon_points*, returns the intersecting fuzzy points of two fuzzy line objects. We specify its signature as $fline \times fline \times itype \rightarrow fpoint$. Let $\tilde{L}_1, \tilde{L}_2 \in fline$. To capture the shared point locations of $\tilde{L}_1$ and $\tilde{L}_2$, we apply the auxiliary operations *extract* and *project* (Section 4.1) to the support of the intersection of $\tilde{L}_1$ and $\tilde{L}_2$. Similarly to fuzzy geometric intersection, we make use of a t-norm operator to determine the membership degree of the common points. Let $f \in itype$. We define *fcommon_points* as follows:

$$fcommon_points(\tilde{L}_1, \tilde{L}_2, f) = \{(p, \mu_{\tilde{L}_1 \cap_f \tilde{L}_2}(p)) \mid p \in project(extract(supp(\tilde{L}_1 \cap_f \tilde{L}_2), \text{“point”})) \wedge \mu_{\tilde{L}_1 \cap_f \tilde{L}_2}(p) = f(\mu_{\tilde{L}_1}(p), \mu_{\tilde{L}_2}(p))\} \in fpoint$$

Fuzzy Common Boundaries. Intuitively, this operation extracts the shared boundary of two fuzzy region objects. Since the boundary of a fuzzy region object consists of two components, we define two operations named $fcommon_boundaries_c$ and $fcommon_boundaries_u$ respectively to get the common areal parts of the two components. For defining them, we compute the fuzzy geometric intersection of the boundary components of two fuzzy region objects, which are extracted by the functions $fboundary_c$ and $fboundary_u$ (Section 4.2). Let $R_1, R_2 \in fregion$. The operations $fcommon_boundaries_c$ and $fcommon_boundaries_u$ have the same signature $fregion \times fregion \times itype \rightarrow fregion$ and are defined as follows:

$$fcommon_boundaries_c(\tilde{R}_1, \tilde{R}_2, f) = fintersection(fboundary_c(\tilde{R}_1), fboundary_c(\tilde{R}_2), f)$$
$$fcommon_boundaries_u(\tilde{R}_1, \tilde{R}_2, f) = fintersection(fboundary_u(\tilde{R}_1), fboundary_u(\tilde{R}_2), f)$$

Fuzzy Common Frontiers. This operation, named *fcommon_frontiers*, captures the common linear parts of the frontiers of two fuzzy region objects. Its signature is specified as $fregion \times fregion \times itype \rightarrow fline$ and its definition is based on the computation of the fuzzy geometric intersection of the frontiers of the fuzzy region objects, which are obtained by the function *ffrontier* (Section 4.2). We define *fcommon_frontiers* as follows:

$$fcommon_frontiers(\tilde{R}_1, \tilde{R}_2, f) = fintersection(ffrontier(\tilde{R}_1), ffrontier(\tilde{R}_2), f)$$

The formal definitions of the operations *fcommon_boundaries*$_c$, *fcommon_boundaries*$_u$, and *fcommon_frontiers* can be straightforwardly extended to deal with fuzzy line objects. This is due to the use of general geometric operations (i.e., *fboundary*$_c$, *fboundary*$_u$, *ffrontier*, and *fintersection*) in their definitions.

4.5 Membership-based Geometric Operations

In this section, we define fuzzy geometric operations that handle and alter the membership degrees of a fuzzy spatial object by using concepts of concentration, dilation, and normalization defined in fuzzy set theory [36]. This means that these operations do not change the geometric shape of the objects but modify the meaning of the spatial fuzziness represented by the objects. This aspect is useful when processing *linguistic modifiers* in fuzzy spatial inference models [8, 11, 16]. In such models, fuzzy spatial objects are employed in the fuzzy propositions to express expert knowledge in applications. A fuzzy proposition has the format L *is* v where L is a linguistic variable and v is a linguistic value of the scope of L. This means that fuzzy spatial objects are labeled by linguistic values, which characterize different instances or occurrences of a linguistic variable. For instance, we can describe the infection level of trees in our running example by using the fuzzy proposition *infection level is high*. In this case, we obtain a fuzzy point object that represents trees with high infection level.

Concentration. Intuitively, this operation, named *concentration*, reduces the membership degrees of a fuzzy spatial object and returns another object. Let $\mathbb{R}_{>1} = \{x \in \mathbb{R} \mid x > 1\}$. We specify the signature of the operation as $\alpha \times \mathbb{R}_{>1} \rightarrow \alpha$. Let $s \in \mathbb{R}_{>1}$. The membership degrees of a fuzzy spatial object $\tilde{O}$ are reduced by squaring them using s. This operation can be used to represent linguistic modifiers like "very" [36]; usually $s = 2$ is selected. We define *concentration* as follows:

$$concentration(\tilde{O}, s) = \{(p, \mu_{con(\tilde{O})}(p)) \mid p \in \mathbb{R}^2 \wedge \mu_{con(\tilde{O})}(p) = \mu_{\tilde{O}}(p)^s\}$$

Dilation. Differently from the previous operation, the operation *dilation* increases the membership degrees of a fuzzy spatial object and returns another object. We specify its signature as $\alpha \times]0, 1[\rightarrow \alpha$. Let $r \in]0, 1[$. The membership degrees of a fuzzy spatial object $\tilde{O}$ are increased by squaring them using r. This operation can be employed to represent linguistic modifiers like "more or less" [36], which relaxes the meaning of the linguistic value and commonly assumes $r = 0.5$. We define *dilation* as follows:

$$dilation(\tilde{O}, r) = \{(p, \mu_{dil(\tilde{O})}(p)) \mid p \in \mathbb{R}^2 \wedge \mu_{dil(\tilde{O})}(p) = \mu_{\tilde{O}}(p)^r\}$$

Normalization. This operation, named *normalization*, divides the membership degrees by the largest membership degree of $\tilde{O}$ (called *height*) and returns another object. Hence, if the height is less than 1, this operation increases the membership degrees of a fuzzy spatial object and includes a core to the object. Otherwise, this operation does not change the degrees of the object. We specify its signature as $\alpha \rightarrow \alpha$. Let $r \in]0, 1[$. To define this operation, we specify the auxiliary operation *height* with signature $\alpha \rightarrow \mathbb{R}$. It is defined as $height(\tilde{O}) = sup\{\mu_{\tilde{O}}(p) \mid p \in \mathbb{R}^2\}$. We define *normalization* as follows:

$$normalization(\tilde{O}) = \{(p, \mu_{norm(\tilde{O})}(p)) \mid p \in \mathbb{R}^2 \wedge \mu_{norm(\tilde{O})}(p) = \mu_{\tilde{O}}(p)/height(\tilde{O})\}$$

4.6 Application to Running Example

In Sections 3.5.3, 3.6.3, and 3.7.3 we have seen how spatial phenomena of our running example are represented by fuzzy spatial data types, which are in turn embedded in relational table schemas as attribute data types. By applying the concept of ADTs (Section 3.4), we can also embed fuzzy spatial operations into an extension of the Structured Query Language (SQL), which is the standard

query language for object-relational database systems. In this section, we illustrate the applicability of FUSA by posing some queries based on our running example. Recall that we have the tables *itree* and *trap* storing fuzzy point objects, *polluted_river* and *animal* keeping fuzzy line objects, and *habitat* and *air_pollution* maintaining fuzzy region objects.

The first query asks for the total course of polluted rivers as the union of all fuzzy region objects representing the course of polluted rivers. For this purpose, we introduce a geometric aggregate function named *agg_funion* that iteratively calls *funion* for each pair of polluted rivers.

```
SELECT agg_funion(P.course)
FROM   polluted_river as P
```

The following query retrieves shared areas between habitats of species and air polluted areas. The particular areas of interest are those with membership degrees greater than or equal to 0.6. Such areas denote locations that need special attention. To this end, we make use of the operation *alpha_cut* and the fuzzy geometric set operation *fintersection*. Further, we are only interested in non-empty areas as indicated by the *where* clause, where *empty_fp*, *empty_fl*, and *empty_fr* denote the empty fuzzy point object $\tilde{\perp}_p$, the empty fuzzy line object $\tilde{\perp}_l$, and the empty fuzzy region object $\tilde{\perp}_r$, respectively.

```
SELECT P.id, A.id, alpha_cut(fintersection(H.habitat, A.area), 0.6)
FROM   habitat as H, air_pollution as A
WHERE  fintersection(H.habitat, A.area) != empty_fr
```

The next query determines the point locations of the insect traps that reached their maximum capacity. It makes use of the operation *core* since it captures the crisp locations with membership degree equal to 1.

```
SELECT T.type, T.when, core(T.tp_loc)
FROM   trap as T
```

The following query finds out the common point locations between animal routes and polluted rivers. The membership degree of the common locations are determined by the *product t-norm* T_{ap} (Section 4.3.1). Such locations are stored in the new attribute *crossing*.

```
SELECT P.id, A.id, fcommon_points(P.course, A.animal_route, T_ap) as crossing
FROM   polluted_river as P, animal as A
WHERE  fcommon_points(P.course, A.animal_route, T_ap) != empty_fp
```

We now ask for the names of species and the areas where their habitats possibly intersect by considering their boundaries only. To this end, we employ the operation $fcommon_boundaries_u$, which returns a fuzzy region object.

```
SELECT A.species, B.species, fcommon_boundaries_u(A.habitat, B.habitat)
       as shared_boundaries
FROM   habitat A, habitat B
WHERE  fcommon_boundaries_u(A.habitat, B.habitat) != empty_fr
```

The next query computes the fuzzy point object that represents the fuzzy proposition *infection level is very high*. For this, we apply the operation *concentration* to all trees with high infection level, which are in turn captured by employing the geometric aggregate function *agg_funion*.

```
SELECT concentration(agg_funion(I.inf_tree), 2)
FROM   itree as I
WHERE  level = 'high'
```

Similarly to the previous query, the final query calculates the fuzzy point object for the proposition *infection level is more or less high* by applying the operation *dilation* to all trees with high infection level.

```
SELECT dilation(agg_funion(I.inf_tree), 0.5)
FROM   itree as I
WHERE  level = 'high'
```

5 Conclusions and Future Work

This article has introduced the *Fuzzy Spatial Algebra* (*FUSA*) as a conceptual and formal foundation for the design and later implementation of *fuzzy spatial data types* and their operations that are useful extensions to spatial database systems and Geographic Information Systems. In this sense, FUSA is a contribution to *spatial information theory*. Its formal framework is based on fuzzy set theory and fuzzy topology. A special feature of FUSA is that users may exert influence on how spatial fuzziness is interpreted and handled in the operations.

We see future work in the three directions of applicability, extensibility, and implementation of FUSA. As for the applicability of FUSA, possible topics are the creation of fuzzy spatial objects from real world datasets (we have performed some initial work in [13]), the use of fuzzy spatial data types in *fuzzy spatial inference models* (we have already contributed to this topic in [8, 11, 16]), and the extraction of spatial association rules from fuzzy spatial data (we have presented some initial work in [26]). As for the extensibility of FUSA, *heterogeneously structured fuzzy spatial collections and compositions* are of interest because some operations return objects of different dimensions. We have already carried out some research in [14]. Further, *fuzzy topological relationships* are of importance since they are able to characterize the relative positions of two fuzzy spatial objects (e.g., if they intersect or meet or are disjoint). We have already made two different proposals that are based on the concepts of *membership degree* [9, 17] and *coverage degree* [10]. Fuzzy numerical operations (e.g., the fuzzy area of a fuzzy region object, the fuzzy length of a fuzzy line object, and fuzzy distance between two fuzzy spatial objects) are also an important class of fuzzy spatial operations that will be covered in future work. In particular, we use them for defining fuzzy topological relationships. Moreover, the handling of three-dimensional fuzzy spatial objects is also part of our future plan in extending FUSA. As for the implementation of FUSA, sophisticated data structures for the fuzzy spatial data types and efficient algorithms for the fuzzy spatial operations and predicates are needed. We have introduced the *Spatial Plateau Algebra* [12, 70] as a possible implementation concept of FUSA and provided *fsr* [7, 18] as its implementation in R.

References

[1] D. Altman. 1994. Fuzzy set theoretic approaches for handling imprecision in spatial analysis. *Int. Journal of Geographical Information Systems* 8, 3 (1994), 271–289.

[2] M. Asadi, M. H. Bagheripour, and M. Eftekhari. 2013. Development of optimal fuzzy models for predicting the strength of intact rocks. *Computers & Geosciences* 54 (2013), 107–112.

[3] L. Bejaoui, F. Pinet, Y. Bedard, and M. Schneider. 2009. Qualified topological relations between spatial objects with possibly vague shape. *Int. Journal of Geographical Information Science* 23, 7 (2009), 877–921.

[4] P. A. Burrough. 1989. Fuzzy mathematical methods for soil survey and land evaluation. *Journal of Soil Science* 40, 3 (1989), 477–492.

[5] P. A. Burrough. 1996. *Natural Objects with Indeterminate Boundaries*, 3–28. In Burrough and Frank [6].

[6] P. A. Burrough and A. U. Frank (Eds.). 1996. *Geographic Objects with Indeterminate Boundaries*. Taylor & Francis.

[7] A. C. Carniel, F. Galdino, J. S. Philippsen, and M. Schneider. 2021. Handling fuzzy spatial data in R using the fsr package. In *ACM SIGSPATIAL Int. Conf. on Advances in Geographic Information Systems*. 526–535.

[8] A. C. Carniel, F. Galdino, and M. Schneider. 2022. Evaluating region inference methods by using fuzzy spatial inference models. In *IEEE Int. Conf. on Fuzzy Systems*. 1–8.

[9] A. C. Carniel and M. Schneider. 2016. A conceptual model of fuzzy topological relationships for fuzzy regions. In *IEEE Int. Conf. on Fuzzy Systems*. 2271–2278.

[10] A. C. Carniel and M. Schneider. 2017. Coverage degree-based fuzzy topological relationships for fuzzy regions. In *Int. Conf. on Flexible Query Answering Systems*. 112–123.

[11] A. C. Carniel and M. Schneider. 2017. Fuzzy inference on fuzzy spatial objects (FIFUS) for spatial decision support systems. In *IEEE Int. Conf. on Fuzzy Systems*. 1–6.
[12] A. C. Carniel and M. Schneider. 2018. Spatial plateau algebra: An executable type system for fuzzy spatial data types. In *IEEE Int. Conf. on Fuzzy Systems*. 1–8.
[13] A. C. Carniel and M. Schneider. 2019. A systematic approach to creating fuzzy region objects from real spatial data sets. In *IEEE Int. Conf. on Fuzzy Systems*. 1–6.
[14] A. C. Carniel and M. Schneider. 2020. Spatial data types for heterogeneously structured fuzzy spatial collections and compositions. In *IEEE Int. Conf. on Fuzzy Systems*. 1–8.
[15] A. C. Carniel and M. Schneider. 2021. A survey of fuzzy approaches in spatial data science. In *IEEE Int. Conf. on Fuzzy Systems*. 1–6.
[16] A. C. Carniel, M. Schneider, and R. R. Ciferri. 2015. FIFUS: A rule-based fuzzy inference model for fuzzy spatial objects in spatial databases and GIS. In *ACM SIGSPATIAL Int. Conf. on Advances in Geographic Information Systems*. 83:1–83:4.
[17] A. C. Carniel, M. Schneider, R. R. Ciferri, and C. D. A Ciferri. 2014. Modeling fuzzy topological predicates for fuzzy regions. In *ACM SIGSPATIAL Int. Conf. on Advances in Geographic Information Systems*. 529–532.
[18] A. C. Carniel, P. V. A. B. Venâncio, and M. Schneider. 2023. fsr: An R package for fuzzy spatial data handling. *Transactions in GIS* 27, 3 (2023), 900–927.
[19] C. L. Chang. 1968. Fuzzy topological spaces. *J. Math. Anal. Appl.* 24, 1 (1968), 182–190.
[20] T. Cheng. 2002. Fuzzy objects: Their changes and uncertainties. *Photogrammetric Engineering & Remote Sensing* 68, 1 (2002), 41–49.
[21] E. Clementini. 2005. A model for uncertain lines. *Journal of Visual Languages & Computing* 16, 4 (2005), 271–288.
[22] E. Clementini and P. Di Felice. 1996. *An Algebraic Model for Spatial Objects with Indeterminate Boundaries*, 155–169. In Burrough and Frank [6].
[23] Y. Cohen, A. Cohen, A. Hetzroni, V. Alchanatis, D. Broday, Y. Gazit, and D. Timar. 2008. Spatial decision support system for Medfly control in citrus. *Computers and Electronics in Agriculture* 62, 2 (2008), 107–117.
[24] A. G. Cohn and N. M. Gotts. 1996. *The 'Egg-Yolk' Representation of Regions with Indeterminate Boundaries*, 171–187. In Burrough and Frank [6].
[25] E. Cuchillo-Ibáñez and Juan Tarrés. 1997. On the boundary of fuzzy sets. *Fuzzy Sets and Systems* 89, 1 (1997), 113–119.
[26] H. P. da Silva, T. D. R. Felix, P. V. A. B. de Venâncio, and A. C. Carniel. 2022. Discovery of spatial association rules from fuzzy spatial data. In *Int. Conf. on Conceptual Modeling*. 179–193.
[27] A. Dilo. 2006. *Representation of and Reasoning with Vagueness in Spatial Information: A System for Handling Vague Objects*. Ph. D. Dissertation. International Institute for Geo-information Science & Earth Observation, University of Twente, The Netherlands.
[28] A. Dilo, R. A. de By, and A. Stein. 2007. A system of types and operators for handling vague spatial objects. *Int. Journal of Geographical Information Science* 21, 4 (2007), 397–426.
[29] G. Fusco, M. Caglioni, K. Emsellem, M. Merad, D. Moreno, and C. Voiron-Canicio. 2017. Questions of uncertainty in geography. *Environment and Planning A: Economy and Space* (2017).
[30] M. F. Goodchild, D. R. Montello, P. Fohl, and J. Gottsegen. 1998. Fuzzy spatial queries in digital spatial data libraries. In *IEEE Int. Conf. on Fuzzy Systems*. 205–210.
[31] T. R. Gottwald. 2010. Current epidemiological understanding of citrus Huanglongbing. *Annual Review of Phytopathology* 48, 1 (2010), 119–139.
[32] T. H. Grubesic. 2006. On the application of fuzzy clustering for crime hot spot detection. *Journal of Quantitative Criminology* 22, 1 (2006), 77–105.
[33] Jifa Guo and Shihong Du. 2021. Building fuzzy areal geographical objects from point sets. *Transactions in GIS* 25, 6 (2021), 3067–3087.
[34] M. S. Islam, M. K. Ahmed, M. Raknuzzaman, M. H. Al-Mamun, and M. K. Islam. 2015. Heavy metal pollution in surface water and sediment: A preliminary assessment of an urban river in a developing country. *Ecological Indicators* 48 (2015), 282–291.
[35] A. Jadidi, M. A. Mostafavi, Y. Bédard, and K. Shahriari. 2014. Spatial representation of coastal risk: A fuzzy approach to deal with uncertainty. *ISPRS International Journal of Geo-Information* 3, 3 (2014), 1077–1100.
[36] M. Jamshidi, N. Vadiee, and T. J. Ross. 1993. *Fuzzy Logic and Control*. Prentice-Hall. 297 pages.
[37] Subhankar Jana and Juthika Mahanta. 2022. Boundary of a fuzzy set and its application in GIS: A review. *Artificial Intelligence Review* (2022), 1–31.
[38] G. Jifa and C. Tiejun. 2014. Geometric properties of interval Type-II fuzzy regions. *Journal of Intelligent and Fuzzy Systems* 26, 2 (2014), 563–575.
[39] R. Kays, M. C. Crofoot, W. Jetz, and M. Wikelski. 2015. Terrestrial animal tracking as an eye on life and planet. *Science* 348, 6240 (2015), 1222.

[40] B. Khalfi, C. de Runz, S. Faiz, and H. Akdag. 2021. A new methodology for storing consistent fuzzy geospatial data in big data environment. *IEEE Transactions on Big Data* 7, 02 (2021), 468–482.

[41] E. P. Klement, R. Meslar, and E. Pap. 2000. *Triangular Norms*. Springer.

[42] R. Krishnapuram, J. M. Keller, and Y. Ma. 1993. Quantitative analysis of properties and spatial relations of fuzzy image regions. *IEEE Trans. on Fuzzy Systems* 1, 3 (1993), 222–233.

[43] P. Lagacherie, P. Andrieux, and R. Bouzigues. 1996. *Fuzziness and Uncertainty of Soil Boundaries: From Reality to Coding in GIS*, 275–286. In Burrough and Frank [6].

[44] Y. Leung and J. Yan. 1997. Point-in-polygon analysis under certainty and uncertainty. *GeoInformatica* 1 (1997), 93–114.

[45] K. Liu and W. Shi. 2006. Computing the fuzzy topological relations of spatial objects based on induced fuzzy topology. *Int. Journal of Geographical Information Science* 20, 8 (2006), 857–883.

[46] K. Liu and W. Shi. 2009. Quantitative fuzzy topological relations of spatial objects by induced fuzzy topology. *Int. Journal of Applied Earth Observation and Geoinformation* 11, 1 (2009), 38–45.

[47] Y. M. Liu and M. K. Luo. 1998. *Advances in Fuzzy Systems - Applications and Theory. Fuzzy Topology*. Vol. 9. World Scientific.

[48] Z. Liu and R. George. 2005. Mining weather data using fuzzy cluster analysis. In *Fuzzy Modeling with Spatial Information for Geographic Problems*, F. E. Petry, V. B. Robinson, and M. A. Cobb (Eds.). Springer-Verlag, 103–119.

[49] W. Lodwick. 2007. *Fuzzy Surfaces in GIS and Geographical Analysis: Theory, Analytical Methods, Algorithms and Applications*. CRC Press.

[50] Xuying Ma, Ian Longley, Jay Gao, Ayushi Kachhara, and Jennifer Salmond. 2019. A site-optimised multi-scale GIS based land use regression model for simulating local scale patterns in air pollution. *Science of The Total Environment* 685 (2019), 134–149.

[51] M. Di Marco, J. E. M. Watson, H. P. Possingham, and O. Venter. 2016. Limitations and trade-offs in the use of species distribution maps for protected area planning. *Journal of Applied Ecology* 54, 2 (2016), 402–411.

[52] M. Molenaar and T. Cheng. 2000. Fuzzy spatial objects and their dynamics. *ISPRS Journal of Photogrammetry & Remote Sensing* 55, 3 (2000), 164–175.

[53] J. Munkres. 2000. *Topology*. Vol. 2. Pearson.

[54] M. M. Noble, D. Harasti, J. Pittock, and B. Doran. 2021. Using GIS fuzzy-set modelling to integrate social-ecological data to support overall resilience in marine protected area spatial planning: A case study. *Ocean & Coastal Management* 212 (2021), 105745.

[55] OGC 2010. OpenGIS Implementation Specification for Geographic Information - Simple Feature Access - Part 2: SQL Option. Open Geospatial Consortium. https://www.ogc.org/standards/sfs

[56] OGC 2011. OpenGIS Implementation Standard for Geographic Information - Simple Feature Access - Part 1: Common Architecture. Open Geospatial Consortium. https://www.ogc.org/standards/sfa

[57] K. Oh and Y. Jeong. 2002. The usefulness of the GIS - fuzzy set approach in evaluating the urban residential environment. *Environment and Planning B: Planning and Design* 29 (2002), 589–606.

[58] P. Pao-Ming and L. Ying-Ming. 1980. Fuzzy topology. I. Neighborhood structure of a fuzzy point and Moore-Smith convergence. *J. Math. Anal. Appl.* 76, 2 (1980), 571–599.

[59] S. Parsons. 1996. Current approaches to handling imperfect information in data and knowledge bases. *IEEE Trans. on Knowledge and Data Engineering* 8, 3 (1996), 353–372.

[60] A. Pauly and M. Schneider. 2010. VASA: An algebra for vague spatial data in databases. *Information Systems* 35, 1 (2010), 111–138.

[61] Z. Pawlak. 1982. Rough sets. *Int. Journal of Computer and Information Sciences* 11, 5 (1982), 341–356.

[62] F. Qi and A.-X. Zhu. 2011. Comparing three methods for modeling the uncertainty in knowledge discovery from area-class soil maps. *Computers & Geosciences* 37, 9 (2011), 1425–1436.

[63] V. B. Robinson. 2003. A perspective on the fundamentals of fuzzy sets and their use in geographic information systems. *Transactions in GIS* 7, 1 (2003), 3–30.

[64] V. B. Robinson. 2007. Issues and challenges of incorporating fuzzy sets in ecological modeling. In *Geographic Uncertainty in Environmental Security*, A. Morris and S. Kokhan (Eds.). Springer-Verlag, 33–51.

[65] R. Rodríguez, P. Gauthier-Maradei, and H. Escalante. 2017. Fuzzy spatial decision tool to rank suitable sites for allocation of bioenergy plants based on crop residue. *Biomass and Bioenergy* 100 (2017), 17–30.

[66] A. Schmitz and A. Morris. 2006. Modeling and manipulating fuzzy regions: Strategies to define the topological relation between two fuzzy regions. *Control and Cybernetics* 35, 1 (2006), 73–95.

[67] M. Schneider. 1997. *Spatial Data Types for Database Systems - Finite Resolution Geometry for Geographic Information Systems*. Vol. LNCS 1288. Springer-Verlag.

[68] M. Schneider. 1999. Uncertainty management for spatial data in databases: Fuzzy spatial data types. In *6th Int. Symp. on Advances in Spatial Databases (LNCS 1651)*. Springer-Verlag, 330–351.

[69] M. Schneider. 2000. Metric operations on fuzzy spatial objects in databases. In *ACM Symp. on Advances in Geographic Information Systems*. 21–26.
[70] M. Schneider. 2014. Spatial plateau algebra for implementing fuzzy spatial objects in databases and GIS: Spatial plateau data types and operations. *Applied Soft Computing* 16, 3 (2014), 148–170.
[71] M. Schneider and T. Behr. 2006. Topological relationships between complex spatial objects. *ACM Trans. on Database Systems* 31, 1 (2006), 39–81.
[72] T. Semeraro, G. Mastroleo, R. Aretano, G. Facchinetti, G. Zurlini, and I. Petrosillo. 2016. GIS fuzzy expert system for the assessment of ecosystems vulnerability to fire in managing Mediterranean natural protected areas. *Journal of Environmental Management* 168 (2016), 94–103.
[73] M. Stonebraker, B. Rubenstein, and A. Guttman. 1983. Application of abstract data types and abstract indices to CAD databases. In *ACM/IEEE Conf. on Engineering Design Applications*. 107–113.
[74] D. Z. Sui. 1992. A fuzzy GIS modeling approach for urban land evaluation. *Computers, Environment and Urban Systems* 16, 2 (1992), 101–115.
[75] X. Tang. 2004. *Spatial Object Modeling in Fuzzy Topological Spaces with Applications to Land Cover Change*. Ph. D. Dissertation. International Institute for Geo-information Science & Earth Observation, University of Twente, The Netherlands.
[76] X. M. Tang, W. Kainz, and Y. Fang. 2005. Reasoning about changes of land covers with fuzzy settings. *Int. Journal of Remote Sensing* 26, 14 (2005), 3025–3046.
[77] E. L. Usery. 1996. *A Conceptual Framework and Fuzzy Set Implementation for Geographic Features*, 71–85. In Burrough and Frank [6].
[78] J. Verstraete. 2009. Fuzzy regions: Interpretations of surface area and distance. *Control and Cybernetics* 38, 2 (2009), 509–526.
[79] J. Verstraete. 2011. Higher reasoning with level-2 fuzzy regions. In *Int. Conf. on Flexible Query Answering Systems*. 49–59.
[80] J. Verstraete. 2012. Deriving topological concepts for fuzzy regions: From properties to definitions. *Control and Cybernetics* 41, 1 (2012), 113–144.
[81] J. Verstraete, G. De Tré, and A. Hallez. 2006. Bitmap based structures for the modeling of fuzzy entities. *Control and Cybernetics* 35, 1 (2006), 147–164.
[82] F. Wang and G. B. Hall. 1996. Fuzzy representation of geographical boundaries in GIS. *Int. Journal of Geographical Information Systems* 10, 5 (1996), 573–590.
[83] R. H. Warren. 1977. Boundary of a fuzzy set. *Indiana University Mathematics Journal* 26, 2 (1977), 191–197.
[84] M. D. Weiss. 1975. Fixed points, separation, and induced topologies for fuzzy sets. *J. Math. Anal. Appl.* 50 (1975), 142–150.
[85] Q.-Y. Yu and D.-Y. Liu. 2003. Modeling region topological relations based on fuzzy spatial data. In *Int. Conf. on Machine Learning and Cybernetics*. 2624–2628.
[86] L. A. Zadeh. 1965. Fuzzy sets. *Information and Control* 8 (1965), 338–353.
[87] F. B. Zhan. 1998. Approximate analysis of binary topological relations between geographic regions with indeterminate boundaries. *Soft Computing* 2, 2 (1998), 28–34.
[88] K. Zheng, X. Zhou, P. C. Fung, and K. Xie. 2012. Spatial query processing for fuzzy objects. *VLDB Journal* 21, 5 (2012), 729–751.

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